

CHAPTER 8

WORK, ENERGY & POWER

1.7 Energy, work and power

1.7.1 Energy

- 1 State that energy may be stored as kinetic, gravitational potential, chemical, elastic (strain), nuclear, electrostatic and internal (thermal)
- 2 Describe how energy is transferred between stores during events and processes, including examples of transfer by forces (mechanical work done), electrical currents (electrical work done), heating, and by electromagnetic, sound and other waves
- 3 Know the principle of the conservation of energy and apply this principle to the transfer of energy between stores during events and processes
- 4 Recall and use the equation for kinetic energy

$$E_k = \frac{1}{2}mv^2$$

- 5 Recall and use the equation for the change in gravitational potential energy

$$\Delta E_p = mg\Delta h$$

1.7.2 Work

- 1 Recall and use the equation
work done = force $\times$ distance moved in the direction of the force
 $W = Fd$

1.7.3 Energy resources

- 1 List renewable and non-renewable energy sources
- 2 Describe how useful energy may be obtained, or electrical power generated, from:
 - (a) chemical energy stored in fossil fuels
 - (b) chemical energy stored in biofuels
 - (c) hydroelectric resources
 - (d) solar radiation
 - (e) nuclear fuel
 - (f) geothermal resources
 - (g) wind
 - (h) tides
 - (i) waves in the seaincluding references to a boiler, turbine and generator where they are used
- 3 Describe advantages and disadvantages of each method limited to whether it is renewable, when and whether it is available, and its impact on the environment

1.7.4 Efficiency

- 1 Define efficiency as:
 - (a) $(\%) \text{ efficiency} = \frac{(\text{useful energy output})}{(\text{total energy input})} (\times 100\%)$
 - (b) $(\%) \text{ efficiency} = \frac{(\text{useful power output})}{(\text{total power input})} (\times 100\%)$and recall and use these equations

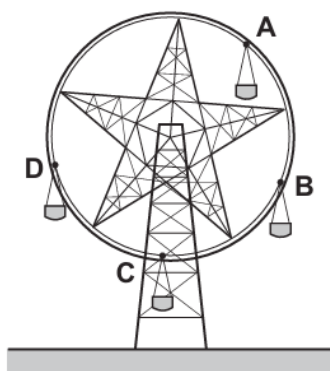
1.7.5 Power

- 1 Define power as work done per unit time and also as energy transferred per unit time; recall and use the equations
 - (a) $\text{power} = \frac{\text{work done}}{\text{time taken}}$
$$P = \frac{W}{t}$$
 - (b) $\text{power} = \frac{\text{energy transferred}}{\text{time taken}}$
$$P = \frac{\Delta E}{t}$$

1. O/N/23/11/Q8

The diagram shows a stationary fairground ride with four chairs of equal mass.

Which chair has the most gravitational potential energy?



2. O/N/23/11/Q9

Hydroelectric and tidal power stations generate electrical energy.

Do these use renewable sources of energy?

	hydroelectric	tidal
A	no	no
B	no	yes
C	yes	no
D	yes	yes

3. O/N/23/11/Q10

Which energy resource is correctly described?

	energy resource	renewable	availability
A	biofuels	yes	at all times
B	nuclear	yes	intermittent
C	solar radiation	no	at all times
D	wind	no	intermittent

4. O/N/23/11/Q11

A small motor has an input power rating of 10 W and is switched on for 5.0 minutes.

What is the electrical energy input to the motor in this time?

- A** 2.0 J **B** 50 J **C** 300 J **D** 3000 J

5. O/N/23/12/Q11

A box with mass 4.0 kg is pulled from rest along a horizontal floor by a constant force F .

A constant frictional force of 1.0 N acts on the box.

When the box has travelled 1.0 m its velocity is 1.5 m/s .

The kinetic energy of the box is equal to the work done on the box.

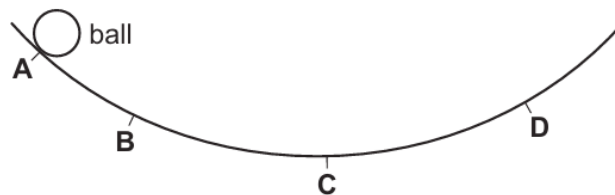
What is force F ?

- A** 2.0 N **B** 3.5 N **C** 4.0 N **D** 5.5 N

6. O/N/23/11/Q12

The diagram shows a curved track. A ball is released from the position shown.

At which point is the gravitational potential energy stored in the ball the maximum?



7. O/N/23/12/Q13

How does an oil-fired power station differ from a nuclear power station?

- A** Gases emitted by hot fuel are emitted into the atmosphere.
B Steam is produced in a boiler using hot fuel.
C The hot steam is used to turn a turbine.
D Turbines are used to drive an electric generator.

8. O/N/23/12/Q15

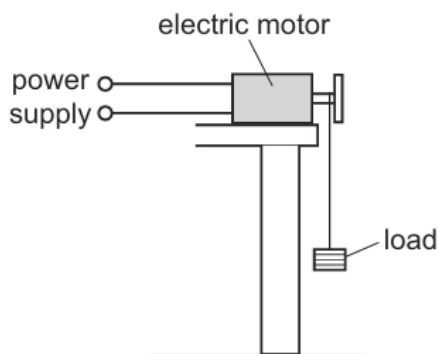
A battery delivers 3000 J of energy to a small 2.0 W electric motor before the battery stops working.

For how long is the motor switched on?

- A** 1500 minutes
B 100 minutes
C 50 minutes
D 25 minutes

9. O/N/23/12/Q14

An electric motor is used to lift a load at a constant speed.



The gravitational potential energy (GPE) stored in the load increases.

The moving load also has stored kinetic energy (KE).

Which expression is used to calculate the efficiency of the motor when the load is being lifted at a constant rate?

A $\text{efficiency} = \frac{\text{GPE gained per second}}{\text{KE gained per second}}$

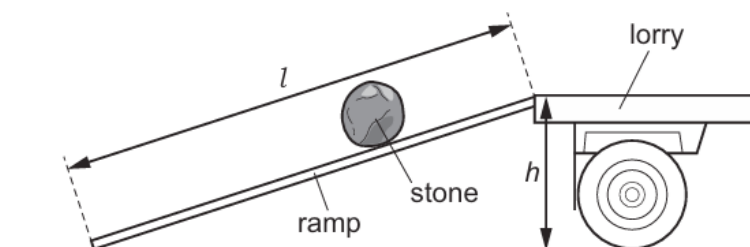
B $\text{efficiency} = \frac{\text{GPE gained per second}}{\text{power supplied}}$

C $\text{efficiency} = \frac{\text{power supplied}}{\text{GPE gained per second}}$

D $\text{efficiency} = \frac{\text{power supplied}}{\text{KE gained per second}}$

10. M/J/23/11/Q11

A large stone is rolled onto a lorry using a ramp.



The length of the ramp is l . The stone is lifted a height of h .

The mass of the stone is m . The weight of the stone is W .

Which expression is equal to the useful work done on the stone?

A $m \times h$

B $W \times h$

C $m \times l$

D $W \times l$

11. M/J/23/11/Q12

The diagram shows a burning candle.



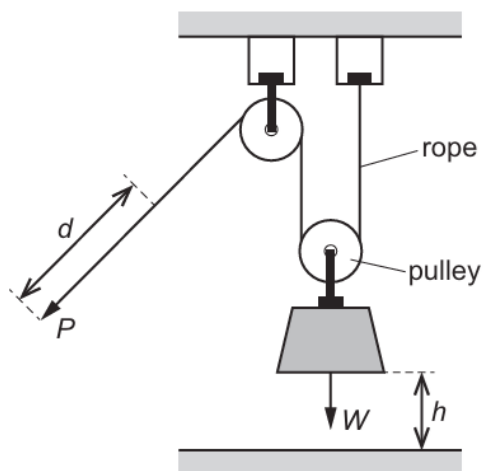
Which energy store is associated with the burning candle?

- A** chemical
- B** elastic
- C** electrostatic
- D** nuclear

12. M/J/23/12/Q12

A worker pulls a rope with a force P through a distance d in the direction of the force as shown.

This causes a block of weight W to move vertically upwards through a height h .



How much work is done by the worker?

- A** Ph
- B** Pd
- C** $(P + W)h$
- D** $(P + W)d$

13. M/J/23/12/Q13

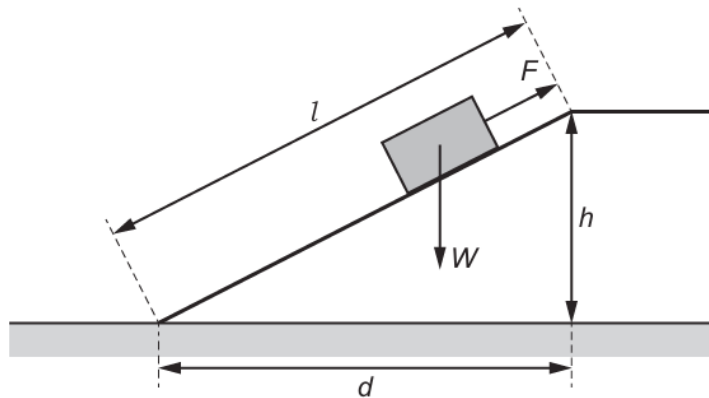
A battery is connected in a circuit to light a lamp.

What is the energy store in the battery?

- A** chemical
- B** electrostatic
- C** gravitational
- D** kinetic

14. O/N/22/11/Q15

A constant force F pulls a block of weight W up the slope shown.



How much work is done by F in pulling the block up the slope?

- A** $F \times h$ **B** $F \times l$ **C** $W \times d$ **D** $W \times l$

15. O/N/22/11/Q16

X and Y are both 40 W, 240 V lamps. Lamp X is more efficient than lamp Y.

Which statement correctly compares X with Y?

- A** X is more expensive to run.
B X has a lower electrical resistance.
C X produces less light energy.
D X produces less thermal energy.

16. O/N/22/12/Q17

The input power to a motor is 12 W.

The motor wastes 590 J of energy in 1.0 min.

What is the efficiency of the motor?

- A** 18% **B** 22% **C** 55% **D** 82%

17. O/N/22/12/Q18

Electric motors have an efficiency of about 90% when used in an electric train.

Which forms of wasted energy are produced?

- A** sound only
B thermal only
C thermal and chemical
D thermal and sound

18. O/N/22/12/Q19

What is a unit of power?

- A** Js **B** J/s **C** Ns **D** N/s

19. M/J/22/11/Q8

A stone falls at terminal velocity in air.

What is equal to the change in gravitational potential energy of the stone?

- A** the change in kinetic energy
B the change in temperature of the stone
C the friction between the stone and the air
D the thermal energy produced

20. M/J/22/11/Q9

Water in a river is used to turn a turbine as the water flows downstream.

How is this type of power source described?

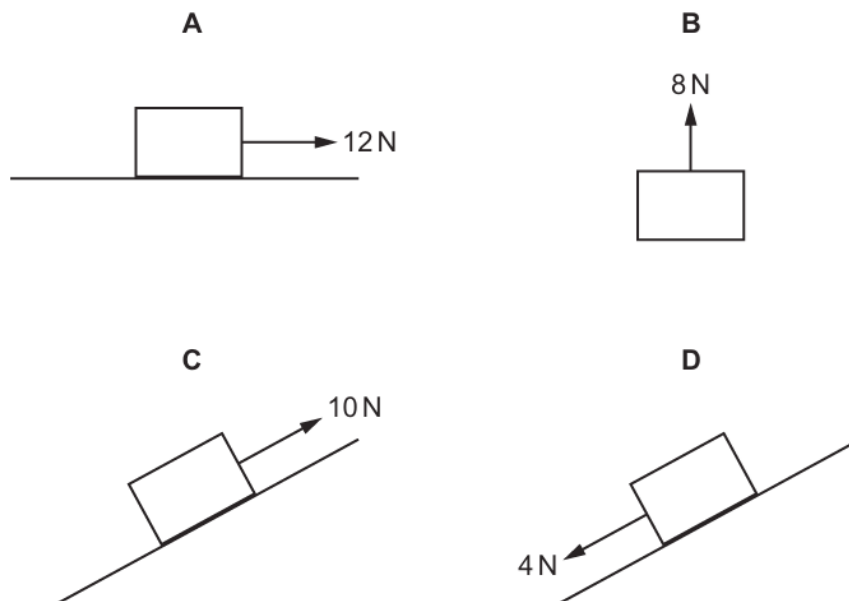
- A** hydroelectric
B tidal
C geothermal
D nuclear

21. M/J/22/11/Q10

A load is pulled by a rope attached to a motor. The resultant force exerted by the rope on the load is shown in the diagrams.

In each diagram, the load moves in the direction of the force shown and takes 10 s to travel 1.0 m.

In which diagram does the motor work with the greatest power?



22. M/J/22/11/Q11

The total energy incident in 1.0 s on a group of solar panels is 120 J/m^2 .

The group of solar panels converts 720 J of light energy to 120 J of useful electrical energy in 1.0 s.

What is the total surface area of the panels and what is the efficiency of the system?

	total surface area of solar panels / m^2	efficiency / %
A	6.0	17
B	0.17	6.0
C	6.0	6.0
D	0.17	17

23. M/J/22/11/Q12

A 1000 W, 240 V electrical device is switched on.

How much energy does it use in 5.0 minutes?

- A** 5000 J **B** 12000 J **C** 72000 J **D** 300000 J

24. M/J/22/12/Q13

Which device provides a continuous, steady energy output for the longest time?

- A** a nuclear reactor
B a solar panel
C a 1.5 V cell
D a wind generator

25. M/J/22/12/Q15

The electrical energy supplied to a kettle is 600 kJ.

Of this energy, 45 kJ is transferred to the surroundings and 15 kJ is used to heat the casing of the kettle.

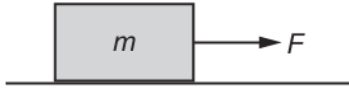
The remaining energy is used to heat the water.

What is the efficiency of the kettle?

- A** 0.10 **B** 0.90 **C** 0.925 **D** 0.975

26. O/N/21/11/Q11

The diagram shows a block of mass m pulled in a straight line along a horizontal surface by a force F .



The block moves a distance d in time t . The average speed at which the block moves is v .

Which two quantities must be known to calculate the work done?

- A** F and d **B** F and m **C** F and t **D** F and v

27. O/N/21/12/Q12

A 60 kg ice-skater, travelling initially at 3.0 m/s, glides across a horizontal icy surface. The skater brakes and comes to rest after 5.0 m.

The work done against the braking force is the change in the kinetic energy of the skater.

What is the braking force?

- A** 18 N **B** 27 N **C** 54 N **D** 108 N

28. O/N/21/12/Q13

Which equation is used to find the efficiency of an energy transfer?

A $\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$

B $\text{efficiency} = \frac{\text{total energy input}}{\text{useful energy output}}$

C $\text{efficiency} = \frac{\text{energy converted to the required form}}{\text{total power input}}$

D $\text{efficiency} = (\text{energy converted to the required form}) \times (\text{total energy input})$

29. M/J/21/11/Q13

The work done by a force F on a body is calculated by multiplying F by a quantity q .

What is q ?

- A** the distance travelled in the direction of the force
B the distance travelled perpendicular to the direction of the force
C the velocity in the direction of the force
D the velocity in the direction perpendicular to the force

30. M/J/21/11/Q14

Some solar panels have a total area of 12 m^2 .

Each 1.0 m^2 of the panels receives 0.85 kJ of energy from the Sun in 1.0 s .

The efficiency of the panels is 16% .

How much power do they produce?

- A** 1.6 kW **B** 2.2 kW **C** 64 kW **D** 160 kW

31. M/J/21/12/Q16

A girl lifts an object of mass 1.0 kg from the floor and puts it on a ledge 2.0 m above the floor.

How much potential energy is gained by the object?

[gravitational field strength $g = 10 \text{ N/kg}$]

- A** 2.0 J **B** 5.0 J **C** 10 J **D** 20 J

32. O/N/20/11/O14

There is a current in a resistor.

Which energy transfer takes place?

- A** electrical $\rightarrow$ chemical
B electrical $\rightarrow$ internal
C kinetic $\rightarrow$ electrical
D potential $\rightarrow$ electrical

33. O/N/20/11/Q16

Which source of electrical energy is **not** renewable?

- A** solar cells
B hydroelectric generators
C wind turbines
D nuclear reactors

34. O/N/20/11/Q17

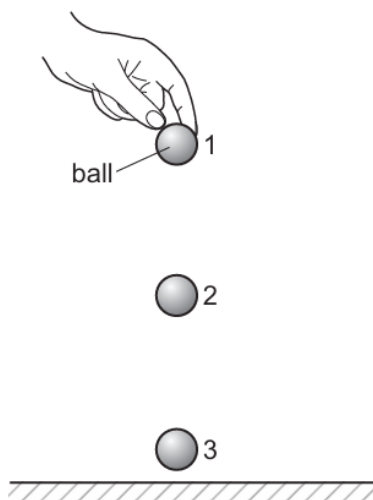
The table shows some data from a high-jump competition.

Which athlete jumps the highest?

	weight / N	increase in gravitational potential energy / J
A	600	1320
B	700	1610
C	800	1760
D	900	1800

35. O/N/20/11/Q15

A ball is dropped from a height as shown.

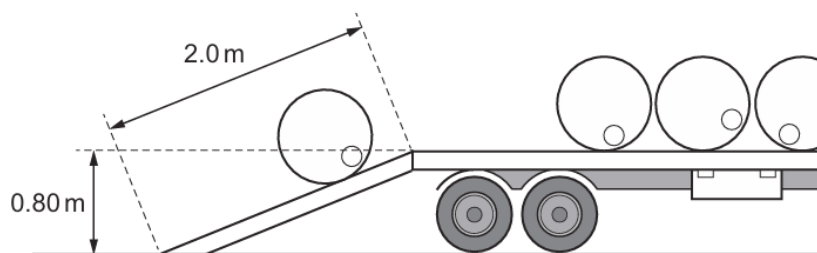


Ignoring the effects of air resistance, which statement about the total energy of the ball is correct?

- A** It is the same at all points.
- B** It is greatest at point 1.
- C** It is greatest at point 2.
- D** It is greatest at point 3.

36. O/N/20/11/Q18

A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

- A** 1000 J
- B** 1600 J
- C** 2500 J
- D** 4000 J

37. O/N/20/11/Q19

Data for three types of electricity generator are shown.

	input energy / MJ	wasted energy / MJ
oil	500	300
nuclear	200	160
hydroelectric	10	1.0

Which is the least efficient generator and which is the most efficient?

	least efficient	most efficient
A	hydroelectric	nuclear
B	hydroelectric	oil
C	nuclear	hydroelectric
D	oil	hydroelectric

38. O/N/20/12/Q14

A student uses an average force of F to push a box a distance of d across a horizontal table. The mass of the box is m and the gravitational field strength is g .

How is the work done by the student pushing the box calculated?

- A** $F \times d$ **B** $\frac{F \times d}{g}$ **C** $m \times d$ **D** $m \times g \times d$

39. O/N/20/12/Q16

What is *power*?

- A** force per second
B the maximum force an object exerts
C the total energy of an object
D work done per second

40. M/J/20/11/Q12

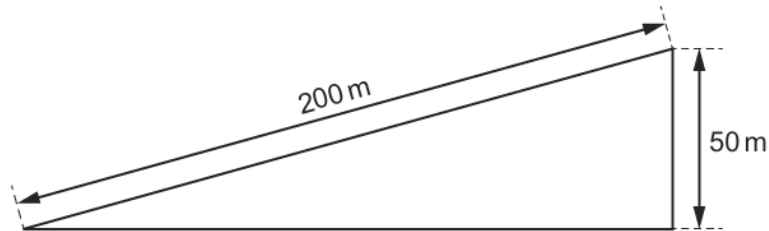
A rocket of mass M when empty carries a mass M of fuel. The rocket and fuel travel at speed v . The engine of the rocket is fired and all of the fuel is expelled. The speed of the rocket increases to $2v$.

What happens to the kinetic energy of the rocket?

- A** It doubles.
B It halves.
C It increases by a factor of four.
D It stays the same.

41. M/J/20/11/Q13

A car of mass 1000 kg is driven 200 m up an incline so that it rises 50 m vertically.



The acceleration of free fall g is 10 m/s^2 .

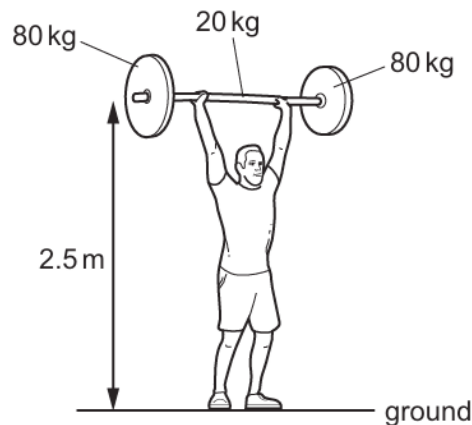
What is the gain in gravitational potential energy?

- A** 5000 J **B** 200 000 J **C** 500 000 J **D** 2 000 000 J

42. M/J/20/11/Q16

In a weightlifting contest, an athlete lifts a metal bar of mass 20 kg fitted with a mass of 80 kg on each end.

The lift from ground to a height of 2.5 m takes 0.50 s.



The gravitational field strength g is 10 N/kg .

What average power does the weightlifter exert in providing the gravitational potential energy during this lift?

- A** 0.90 kW **B** 5.0 kW **C** 8.0 kW **D** 9.0 kW

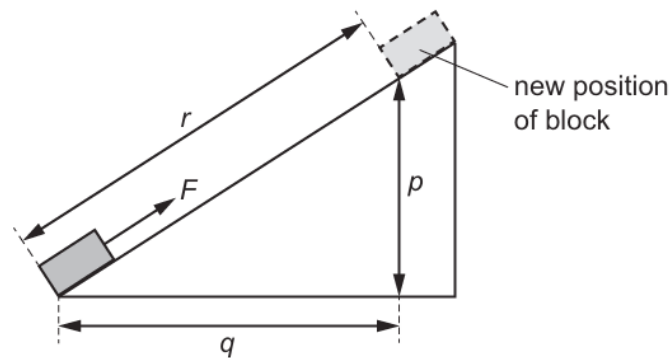
43. M/J/20/11/Q18

In which situation is energy being released by the fusion of hydrogen nuclei to form helium?

- A** in the decay of ^{14}C used to date an object
B in a radioactive isotope emitting alpha-particles
C in the centre of the Earth
D in the centre of the Sun

44. M/J/20/11/Q19

The diagram shows a block being pulled up a slope by a force F .



The block reaches the new position at the top of the slope.

What is the work done by force F in moving the block to its new position?

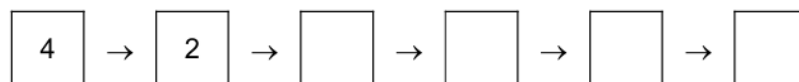
- A** $F \times r$ **B** $F \times p$ **C** $F(q + p)$ **D** $F(r + p)$

45. M/J/20/12/Q19

The main stages in the operation of a coal-fired power station are listed. They are **not** in the correct order.

- 1 The turbine turns a generator.
- 2 Water in the boiler becomes hot.
- 3 Steam turns a turbine.
- 4 Burning coal produces thermal energy.
- 5 Electromagnetic induction produces electrical energy.
- 6 Steam is produced.

The flow chart shows the first two stages.



What is the correct order for the remaining stages?

- A** $6 \rightarrow 3 \rightarrow 1 \rightarrow 5$
B $6 \rightarrow 1 \rightarrow 3 \rightarrow 5$
C $1 \rightarrow 6 \rightarrow 3 \rightarrow 5$
D $6 \rightarrow 3 \rightarrow 5 \rightarrow 1$

46. M/J/20/12/Q20

In which situation is there **no** transfer of energy?

- A a car battery turning the starter motor
- B a car moving along a horizontal road at constant speed
- C a solar panel warming water
- D a spacecraft orbiting the earth at constant height

47. M/J/20/12/Q21

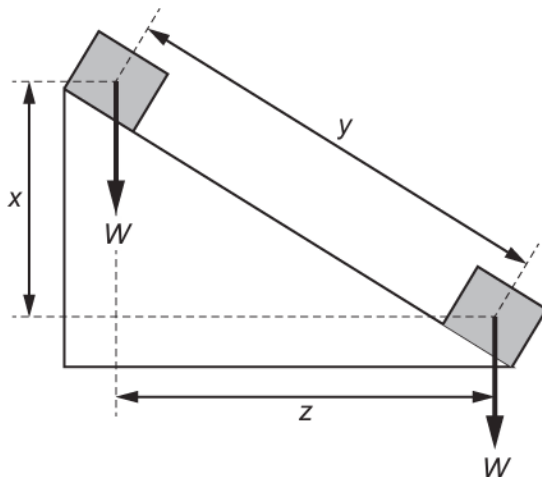
The 200 V battery of an electric car is charged with a current of 33 A for 4.0 hours. The efficiency of the charging process is 90%.

How much useable energy is transferred to the battery?

- A 21 MJ B 86 MJ C 95 MJ D 106 MJ

48. O/N/19/11/Q17

A heavy box of weight W slides down a frictionless slope.



What is the work done by the weight W in moving from top to bottom?

- A Wx B Wy C $W(y - z)$ D $W(x + z)$

49. O/N/19/11/Q18

Where is energy released by the fusion of hydrogen nuclei to form helium?

- A in a nuclear power station
- B in a radioactive isotope emitting alpha-particles
- C in the core of the Earth
- D in the core of the Sun

50. O/N/19/12/Q14

An object X has mass m and velocity v .

A second object Y has mass $2m$ and velocity $2v$.

What is the value of the ratio $\frac{\text{kinetic energy of Y}}{\text{kinetic energy of X}}$?

- A** 1 **B** 2 **C** 4 **D** 8

51. O/N/19/12/Q16

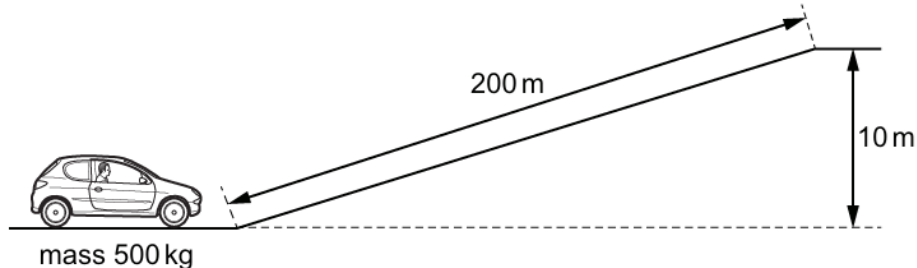
A student uses a newton meter to pull an object a distance d horizontally along a flat surface in a time of t . The newton meter reading is F .

What is given by the expression $(F \times d)/t$?

- A** efficiency
B energy
C power
D work

52. M/J/19/11/Q15

The diagram shows a small car of mass 500 kg approaching a hill. It moves up the hill with constant speed.



The gravitational field strength g is 10 N/kg .

Ignoring friction, how much work is done in moving the car up the hill?

- A** $5.0 \times 10^3 \text{ J}$ **B** $5.0 \times 10^4 \text{ J}$ **C** $1.0 \times 10^5 \text{ J}$ **D** $1.0 \times 10^6 \text{ J}$

53. M/J/19/12/Q16

An object is lifted.

Which expression gives the increase in gravitational potential energy of the object?

- A** mass multiplied by its velocity
B mass multiplied by its acceleration
C the rate at which work is done
D weight multiplied by the increase in height

54. M/J/19/12/Q18

Which method of producing electricity does **not** use an a.c. generator?

- A** geothermal power station
- B** solar panels
- C** wave-energy generators
- D** wind turbine

55. M/J/19/12/Q19

Three solar panels have a total area of 7.2 m^2 . Together, they produce 1.0 kW of power.

An area of 1.0 m^2 of each panel receives 0.80 kJ of energy from the Sun in 1.0 s .

What is the efficiency of the solar panels?

- A** 11% **B** 17% **C** 52% **D** 80%

56. M/J/19/12/Q20

A student suggests three different definitions for power.

- 1 energy transferred per second
- 2 force $\times$ distance moved
- 3 rate of doing work

Which definitions are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 3 only

57. O/N/18/11/Q9

An electric motor, connected to the mains, is used to lift bricks to the top of a building.

There is a fuse in the plug.

How can the efficiency of the motor be increased?

- A** Increase the friction in the motor.
- B** Reduce the energy losses in the motor.
- C** Use a fuse with a higher current rating.
- D** Use a fuse with a lower current rating.

58. O/N/18/12/Q8

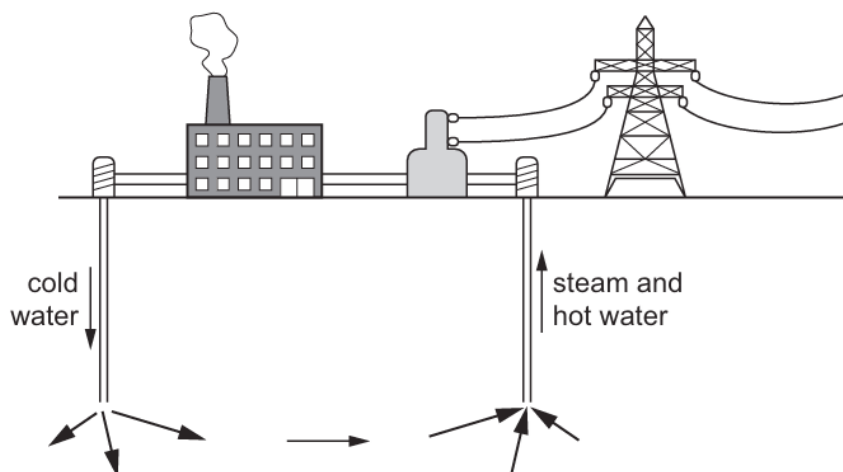
A tennis ball is compressed. Work is done and energy is transferred to the molecules of air inside the ball, and also to the rubber from which the ball is made.

Which row is correct?

	kinetic energy of air molecules inside the ball	temperature of ball
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

59. O/N/18/12/O11

The diagram represents a geothermal power station.

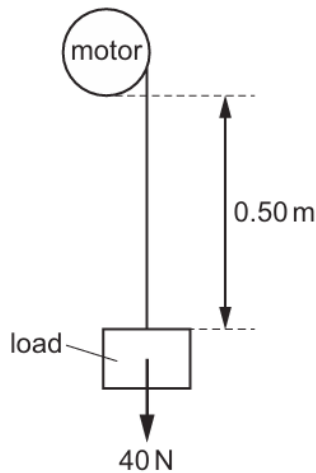


Which useful energy transformation is taking place?

- A** electrical energy → potential energy
- B** electrical energy → thermal energy
- C** potential energy → electrical energy
- D** thermal energy → electrical energy

60. O/N/18/12/Q12

A motor is used to lift a load 0.50 m vertically, as shown.



The load weighs 40 N. The power of the motor is 20 W and the system is 25% efficient.

How long does it take to raise the load?

- A** 0.040 s **B** 0.25 s **C** 4.0 s **D** 40 s

61. M/J/18/11/Q12

A student calculates his power in running up a flight of stairs. He measures the vertical height of the stairs, the time taken to run up the stairs and his weight.

How does he calculate his power?

- A** height $\times$ time $\times$ weight
B $\frac{\text{height} \times \text{weight}}{\text{time}}$
C $\frac{\text{time} \times \text{weight}}{\text{height}}$
D $\frac{\text{weight}}{\text{height} \times \text{time}}$

62. M/J/18/11/Q13

How does an oil-fired power station differ from a nuclear power station?

- A** Gases emitted by hot fuel are emitted into the atmosphere.
B Steam is produced in a boiler using hot fuel.
C The hot steam is used to turn a turbine.
D Turbines are used to drive an electric generator.

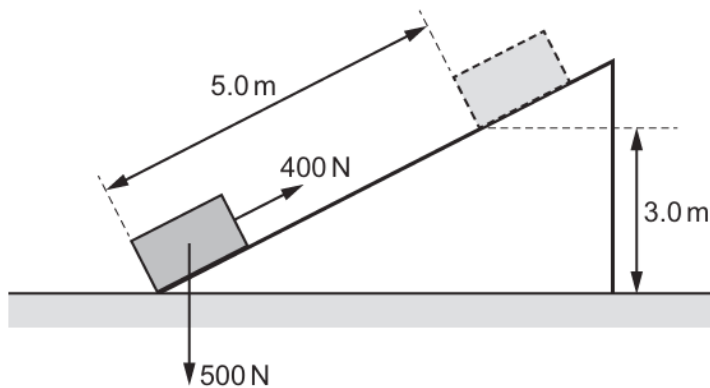
63. M/J/18/12/Q14

Which power station produces carbon dioxide when operating?

- A** gas-fired power station
- B** geothermal power station
- C** nuclear power station
- D** wind power station

64. M/J/18/12/Q15

Work is done when a force of 400 N pulls a crate of weight 500 N at a constant speed along a ramp, as shown.



Part of the work done increases the gravitational potential energy E of the crate and the rest is work done W against friction.

What are the values of E and W ?

	E/J	W/J
A	1500	500
B	1500	2000
C	2000	2500
D	3500	500

65. O/N/17/11/Q9

An astronaut on the Moon drops a tool box of mass 3.0 kg. It falls from rest and its kinetic energy as it hits the surface is 0.96 J.

At which speed does the tool box hit the surface of the Moon?

- A** 0.46 m/s
- B** 0.57 m/s
- C** 0.64 m/s
- D** 0.80 m/s

66. O/N/17/11/Q10

A car travels a distance of 200 m in 20 s. The engine of the car provides a driving force of 1000 N.

What is the power output of the engine?

- A** 0.25 W **B** 4.0 W **C** 100 W **D** 10 000 W

67. O/N/17/12/Q8

A tennis ball of mass 56 g is travelling at 1500 metres / minute.

Which expression is equal to the kinetic energy, in joules, of the tennis ball?

- A** $\frac{1}{2} \times 0.056 \times (25)^2$
B $\frac{1}{2} \times 0.056 \times (1500)^2$
C $\frac{1}{2} \times 56 \times (25)^2$
D $\frac{1}{2} \times 56 \times (1500)^2$

68. O/N/17/12/Q9

In a hydroelectric power station, 4.2×10^5 kg of water passes through the turbines every second. The turbines are at a height of 50 m below the surface of the reservoir. The gravitational field strength g is 10 N/kg.

Assuming there are no energy losses, what is the power output of the power station?

- A** 8.4×10^3 W **B** 8.4×10^4 W **C** 2.1×10^7 W **D** 2.1×10^8 W

69. M/J/17/11/Q15

A small hydroelectric power station diverts water from a river. Every second, 20 kg of water flows through a pipe and falls through a vertical drop of 15 m. The efficiency of the power station is 0.60 (60%).

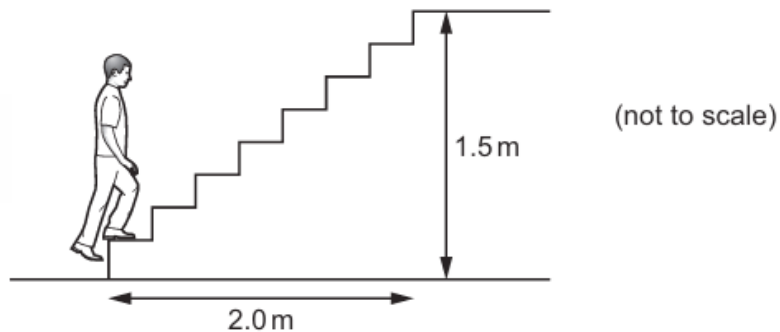
The gravitational field strength g is 10 N/kg.

What is the power output?

- A** 0.18 kW **B** 1.8 kW **C** 3.0 kW **D** 180 kW

70. M/J/17/12/Q15

A student of mass 60 kg climbs some steps. He travels a horizontal distance of 2.0 m and a vertical distance of 1.5 m. The gravitational field strength g is 10 N/kg.



What is the work done against the force of gravity?

- A** 90 J **B** 120 J **C** 900 J **D** 1200 J

71. M/J/17/12/Q16

Which energy source is non-renewable?

- A** nuclear energy
B solar energy
C tidal energy
D wind energy

72. M/J/17/12/Q7

In a coal-fired power station, the coal is burnt and thermal energy (heat) is produced. The thermal energy is used to produce electrical energy.

In which order does the energy pass through parts of the power station?

- A** boiler → generator → turbine
B boiler → turbine → generator
C turbine → boiler → generator
D turbine → generator → boiler

73. O/N/16/11/Q9

Three objects P, Q and R have different masses and different speeds, as shown in the table.

	$\frac{\text{mass}}{\text{kg}}$	$\frac{\text{speed}}{\text{m/s}}$
P	1.0	3.0
Q	2.0	2.0
R	5.0	1.0

What is the order of increasing kinetic energy (smallest first) of the objects?

A P → Q → R

B P → R → Q

C R → P → Q

D R → Q → P

74. O/N/16/11/Q10

A weightlifter lifts some masses through a height of 1.8 m.



The gravitational field strength g is 10 N/kg.

During the lift, the gravitational potential energy of the masses increases by 4500 J.

What is the total mass lifted?

A 25 kg

B 250 kg

C 450 kg

D 810 kg

75. O/N/16/11/Q11

Which expression is used to calculate power?

A $\frac{\text{distance}}{\text{work done}}$

B work done $\times$ distance

C $\frac{\text{work done}}{\text{time taken}}$

D work done $\times$ time taken

76. O/N/16/12/Q12

Which power stations all rely on renewable sources of energy?

- A hydroelectric, oil-fired, solar power stations
- B hydroelectric, solar, wind power stations
- C hydroelectric, oil-fired, wind power stations
- D oil-fired, wind, solar power stations

77. O/N/16/12/Q13

Which force does no work?

- A the force from a car's engine that causes the car to accelerate
- B the force that causes a box to slide over a rough horizontal surface
- C the force that causes a satellite to move in a circular orbit
- D the force that causes a spring to stretch

78. O/N/16/12/Q15

An electric motor is used to operate a lift (elevator). The motor does 140 000 J of work in 70 s.

What is the output power of the motor?

- A 0.50 milliwatts
- B 2.0 watts
- C 2.0 kilowatts
- D 9.8 megawatts

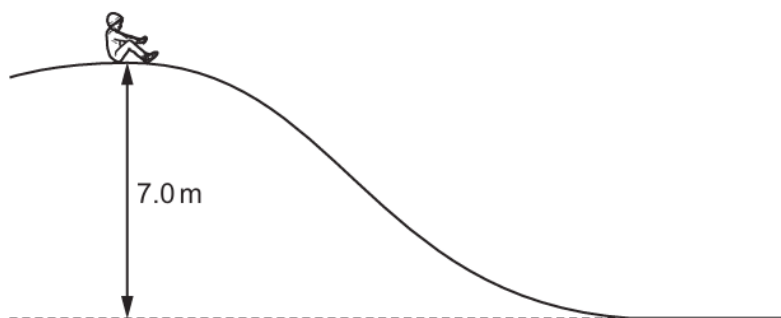
79. O/N/16/12/Q16

Which expression gives the efficiency of an a.c. generator?

- A $\frac{\text{electrical energy output}}{\text{total energy input}}$
- B $\frac{\text{electrical energy output}}{\text{thermal energy output}}$
- C $\frac{\text{total energy output}}{\text{electrical energy input}}$
- D $\frac{\text{thermal energy output}}{\text{total energy input}}$

80. M/J/16/11/Q11

A child slides down a slide.



The weight of the child is 250 N. The height of the slide is 7.0 m. The work done against friction as the child travels down the slide is 1300 J.

What is the change in gravitational potential energy and what is the final kinetic energy of the child?

	change in gravitational potential energy	final kinetic energy
A	1750	450
B	1750	1750
C	17 500	16 200
D	17 500	17 500

81. M/J/16/11/Q12

What uses non-renewable energy?

- A** a geothermal heating system
- B** a nuclear power station
- C** a solar panel
- D** a wind turbine

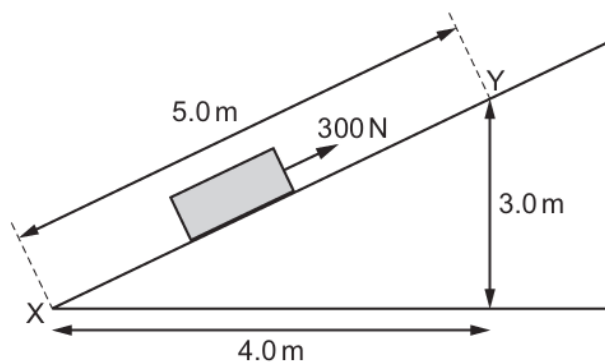
82. M/J/16/11/Q13

Which process in the Sun produces energy?

- A** burning
- B** nuclear fission
- C** nuclear fusion
- D** radiation

83. M/J/16/11/Q14

A 300 N force is applied to a box to move it up a ramp, as shown.



How much work is done by the force when moving the box from X to Y?

- A** 900 J **B** 1200 J **C** 1500 J **D** 3000 J

84. O/N/15/11/Q14

The input power to a lamp is 6.0 W. The lamp wastes 2.7 J of energy in 3.0 s.

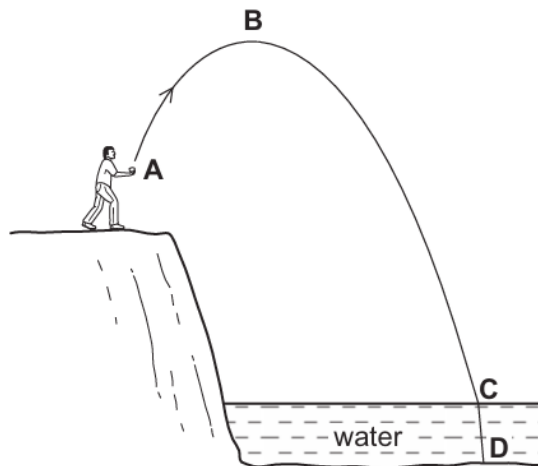
What is the efficiency of the lamp?

- A** 0.15 **B** 0.45 **C** 0.55 **D** 0.85

85. O/N/15/12/Q14

A person throws a stone so that it follows the path shown in the diagram.

In which position does the stone have the most gravitational potential energy?



86. O/N/15/12/Q15

A piston of area 10 cm^2 is pushed slowly into a very large cylinder containing gas at a pressure of 10 N/cm^2 . The pressure of the gas remains constant as the piston moves a distance of 0.10 m .

What is the force of the gas on the piston and what is the work done by the piston on the gas?

	force / N	work done / J
A	1.0	0.1
B	1.0	10
C	100	10
D	100	1000

87. M/J/15/11/Q11

A cyclist travels along a horizontal track at constant speed.

The work done by the cyclist is equal to

- A** the change in kinetic energy.
- B** the force of air resistance.
- C** the force of friction in the bicycle.
- D** the thermal energy (heat) produced.

88. M/J/15/11/Q12

A 2.0 kg mass has 300 J of kinetic energy.

What is the speed of the mass?

- A** 8.7 m/s **B** 12 m/s **C** 17 m/s **D** 24 m/s

89. M/J/15/12/Q12

A rocket of total mass M is travelling at a speed v . The engine of the rocket is fired and fuel is used up. The mass of the rocket decreases to $M/2$ and its speed increases to $2v$.

What happens to the kinetic energy of the rocket?

- A** it doubles
- B** it halves
- C** it increases by a factor of four
- D** it stays the same

90. M/J/15/12/Q13

A builder lifts eight slabs from the ground on to the back of a lorry 1.5 m high.

The total time taken is 48 s and each slab weighs 200 N.

How much useful power does the builder produce?

- A** 50 W **B** 400 W **C** 2400 W **D** 3200 W

91. O/N/14/11/Q13

A student uses a newton meter to pull an object horizontally along rough ground. The student multiplies the newton meter reading by the distance moved by the object.

What does this calculation give?

- A** the efficiency of the process
B the kinetic energy of the object
C the power generated
D the work done on the object

92. O/N/14/11/Q14

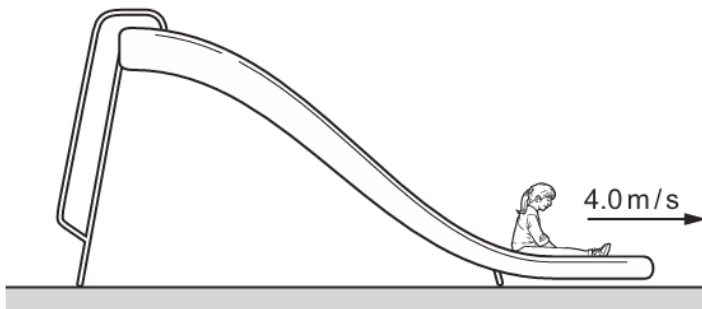
A builder carrying six bricks climbs a ladder. Each brick weighs 25 N and the builder climbs a vertical distance of 4.0 m in 8.0 s.

What is the average power used to raise the bricks?

- A** 13 W **B** 75 W **C** 750 W **D** 4800 W

93. O/N/14/12/Q9

A child of mass 30 kg is moving at a speed of 4.0 m/s when she reaches the bottom of a slide.



What is her kinetic energy?

- A** 60 J **B** 120 J **C** 240 J **D** 480 J

94. M/J/14/11/Q16

A crane lifts a load of 6000 N through a vertical distance of 15 m in 30 s.

What is the average useful power during this operation?

- A** 200 W **B** 400 W **C** 3000 W **D** 12 000 W

95. M/J/14/11/Q17

A javelin has a mass of 0.80 kg.

The javelin is released with 250 J of kinetic energy.

What is the speed of the javelin?

- A** 13 m/s **B** 18 m/s **C** 25 m/s **D** 630 m/s

96. M/J/14/12/Q16

Which source releases carbon dioxide, a greenhouse gas, when generating electricity?

- A** fossil fuels
B geothermal
C hydroelectric
D nuclear

97. O/N/13/11/Q12

Which energy resource comes from hot rocks beneath the Earth's surface?

- A** geothermal energy
B hydroelectric energy
C solar energy
D wind energy

98. O/N/13/12/13

When exposed to light, a solar cell generates electrical energy.

Two solar cells of equal area are tested. Each cell is exposed to sunlight of different brightness for the same time.

Which statement could apply to the more efficient cell?

- A** It generates less electrical energy from light of greater brightness.
B It generates less electrical energy from light of the same brightness.
C It generates the same electrical energy from light of greater brightness.
D It generates the same electrical energy from light of lower brightness.

99. M/J/13/11/Q10

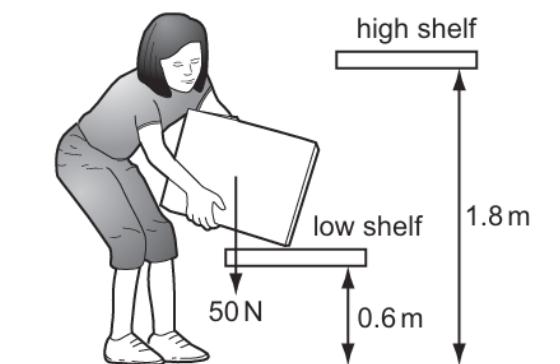
Energy is released in some power stations and in the Sun by either nuclear fission or nuclear fusion.

Which type of nuclear reaction applies in each case?

	power station	Sun
A	fission	fission
B	fission	fusion
C	fusion	fission
D	fusion	fusion

100. M/J/13/11/Q11

The diagram shows a girl lifting a box of weight 50 N from a low shelf to a high shelf.

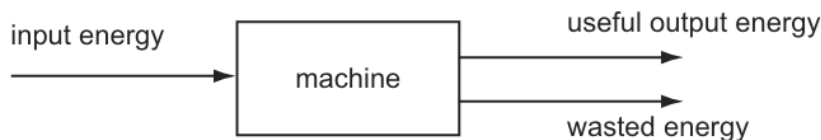


How much work is done on the box?

- A** 30 J **B** 60 J **C** 90 J **D** 120 J

101. M/J/13/11/Q12

The diagram shows the energy transfer through a machine.



The machine is 50% efficient.

Which equation is correct?

- A** input energy = useful output energy
B useful output energy = input energy + wasted energy
C wasted energy = input energy + useful output energy
D wasted energy = useful output energy

102. M/J/13/12/Q13

A swimmer dives into a very deep pool at high speed. He slows down as he moves towards the bottom of the pool.

What is the overall energy transformation as the diver moves downwards through the water?

- A** gravitational potential energy → kinetic energy + thermal energy (heat)
- B** gravitational potential energy → kinetic energy → thermal energy (heat)
- C** kinetic energy + gravitational potential energy → thermal energy (heat)
- D** kinetic energy → gravitational potential energy + thermal energy (heat)

103. M/J/13/12/Q14

A lorry of mass 10 000 kg takes 5000 kg of sand to the top of a hill 50 m high, unloads the sand and then returns to the bottom of the hill.

The gravitational field strength is 10 N/kg.

What is the overall gain in potential energy?

- A** 250 000 J **B** 750 000 J **C** 2 500 000 J **D** 7 500 000 J

104. O/N/12/11/Q13

Which is a non-renewable source of energy?

- A** a battery in a watch
- B** a hydroelectric power station
- C** a solar panel
- D** a wind turbine

105. O/N/12/11/Q14

A builder lifts eight slabs from the ground on to the back of a lorry 1.5 m high.

The total time taken is 48 s and each slab weighs 20 N.

How much useful power does the builder produce?

- A** 5 W **B** 40 W **C** 240 W **D** 320 W

106. O/N/12/12/Q13

Hydroelectric, tidal and fossil fuel power stations generate electrical energy.

Do these use renewable sources of energy?

	hydroelectric	tidal	fossil fuel
A	no	yes	yes
B	no	no	yes
C	yes	no	no
D	yes	yes	no

107. O/N/12/12/Q14

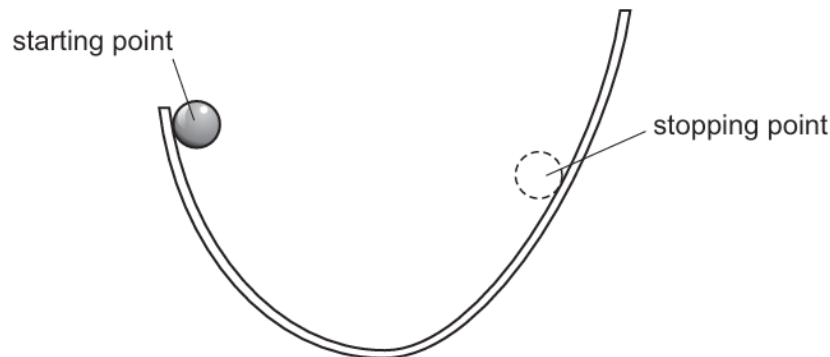
An object of mass 5 kg is carried a distance of 4 m across a room at a constant height above the floor.

What is the work done on the object?

- A** 0 **B** 20 J **C** 50 J **D** 200 J

108. M/J/12/11/Q13

A ball is held at rest on one side of a curved track.



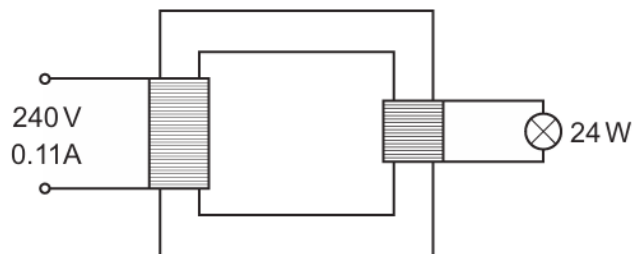
The ball is released. It rolls down one side of the track and part of the way up the other side. It then stops, before rolling back down again. The height of the stopping point is less than that of the starting point.

What is the sequence of energy changes between starting and stopping for the first time?

- A** potential energy → kinetic energy → potential energy
B potential energy → kinetic energy → heat → potential energy
C potential energy → heat → kinetic energy → potential energy
D potential energy → kinetic energy + heat → potential energy + heat

109. M/J/12/11/Q14

A transformer connected to a 240 V mains supply is used to light a 24 W lamp.



The input current to the transformer is 0.11 A and the input voltage is 240 V. The useful output power of the transformer is 24 W.

What is the efficiency of the transformer?

- A** 0.10 **B** 0.91 **C** 1.1 **D** 2.6

110. M/J/12/12/Q13

Two major components of a coal-fired power station are a turbine and a generator.

What are the output forms of energy from the turbine and from the generator?

	turbine	generator
A	electrical	electrical
B	electrical	kinetic
C	heat	kinetic
D	kinetic	electrical

111. M/J/12/12/Q14

What is efficiency?

- A** $\frac{\text{total energy input}}{\text{useful energy output}}$
- B** $\frac{\text{total power input}}{\text{useful energy output}}$
- C** $\frac{\text{useful energy output}}{\text{total energy input}}$
- D** $\frac{\text{useful power output}}{\text{total energy input}}$

112. O/N/11/11/Q10

The efficiency of an electrical generator is 65 %.

The energy input to the generator is 12 kJ.

What is the useful energy output?

- A** 4.2 kJ **B** 5.4 kJ **C** 7.8 kJ **D** 780 kJ

113. O/N/11/11/Q11

Coal is burned as fuel to heat water in a boiler, producing steam. The steam drives a turbine, which is connected to an electric generator.

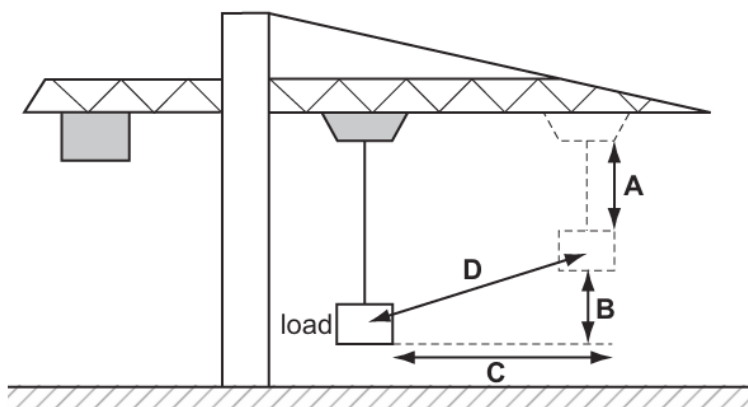
In which order do the major energy transformations take place?

- A** chemical energy → heat energy → electrical energy → kinetic energy
- B** chemical energy → heat energy → kinetic energy → electrical energy
- C** heat energy → chemical energy → electrical energy → kinetic energy
- D** heat energy → chemical energy → kinetic energy → electrical energy

114. O/N/11/11/Q12

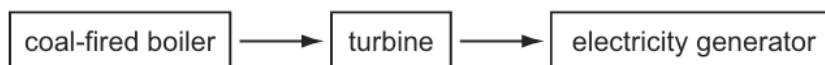
A crane moves its load diagonally, as shown.

By which distance is the weight of the load multiplied to calculate the change in gravitational potential energy of the load?



115. M/J/11/11/Q12

The diagram represents parts of a power station.



What is the order of the energy changes taking place?

- A chemical → heat → kinetic → electrical
- B chemical → kinetic → heat → electrical
- C heat → chemical → kinetic → electrical
- D kinetic → chemical → heat → electrical

116. M/J/11/11/Q13

The centre of the Sun produces large amounts of energy.

What is the source of this energy?

- A chemical reaction
- B nuclear fission
- C nuclear fusion
- D radioactive decay

117. M/J/11/11/Q14

A crane lifts a weight of 1000 N through a vertical height of 30 m.

It uses 60 000 J of energy.

What is the efficiency of the crane?

- A 20 %
- B 30 %
- C 50 %
- D 200 %

118. O/N/10/11/Q15

The energy output of a generator depends on its efficiency and the energy input.

Which set of values is correct?

	efficiency	energy input / MJ	energy output / MJ
A	0.1	50	25
B	0.2	30	6
C	0.4	120	30
D	0.6	60	24

119. M/J/10/11/Q12

A parachutist has opened his parachute and is falling to Earth at constant speed.

What is the principal energy conversion taking place as he falls?

- A** kinetic energy → potential energy
- B** kinetic energy → thermal energy (heat)
- C** potential energy → kinetic energy
- D** potential energy → thermal energy (heat)

120. M./J/10/11/Q13

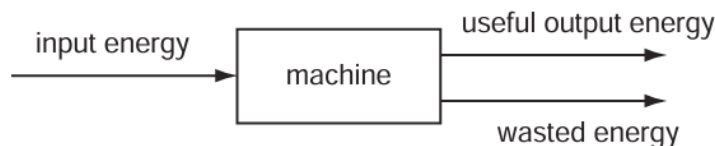
A small emergency generator supplies 432 000 000 J of electrical energy in twenty-four hours.

What is the average power output of the generator?

- A** 5000 W
- B** 300 000 W
- C** 18 000 000 W
- D** 432 000 000 W

121. O/N/09/Q12

The diagram shows energy transfer through a machine.



What is the efficiency of the machine?

- A $\frac{\text{input energy}}{\text{useful output energy}}$
- B $\frac{\text{useful output energy}}{\text{input energy}}$
- C $\frac{\text{useful output energy}}{\text{wasted energy}}$
- D $\frac{\text{wasted energy}}{\text{input energy}}$

122. O/N/09/Q13

A man weighs 600 N. He runs up a staircase of total height 4.0 metres in 3.0 seconds.

How much useful power is needed to do this?

- A 450 W B 800 W C 2400 W D 7200 W

123. M/J/09/Q13

A crane lifts a load of 1000 N through a vertical height of 3.0 m in 10 s. The input power to the crane is 500 J/s.

What is the efficiency of the crane?

- A 0.17 B 0.50 C 0.60 D 0.67

124. O/N/08/Q12

A crane lifts a weight of 1000 N through a vertical height of 30 m.

It uses 60 000 J of energy.

What is the efficiency of the crane?

- A 20 % B 30 % C 40 % D 50 %

125. O/N/08/Q13

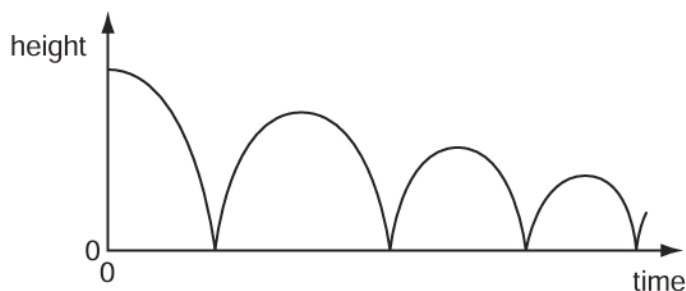
A rock climber of weight 600 N climbs up a rock face of vertical height 300 m in 3600 s.

What is the average power she generates against gravity during this time?

- A 0.020 W B 50 W C 1800 W D 7200 W

126. O/N/08/Q10

The graph shows how the height above the ground of a bouncing ball changes with time.

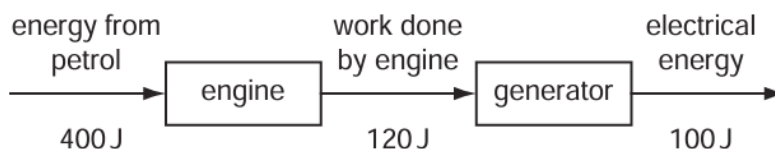


Which statement explains why the height of each peak decreases with time?

- A** Kinetic energy is converted to potential energy at each bounce.
- B** Kinetic energy is converted to thermal energy at each bounce.
- C** The ball gains energy on impact with the floor.
- D** The ball is wearing out.

127. M/J/08/Q11

Energy from petrol is used to operate an engine. The engine drives a generator, which produces electrical energy.



What is the overall efficiency of the process?

- A** 25 %
- B** 30 %
- C** 55 %
- D** 83 %

128. O/N/07/Q9

A 2 kg mass is moving at constant speed.

The kinetic energy of the mass is 400 J.

What is the speed of the mass?

- A** 0.4 m/s
- B** 20 m/s
- C** 200 m/s
- D** 400 m/s

129. O/N/07/Q11

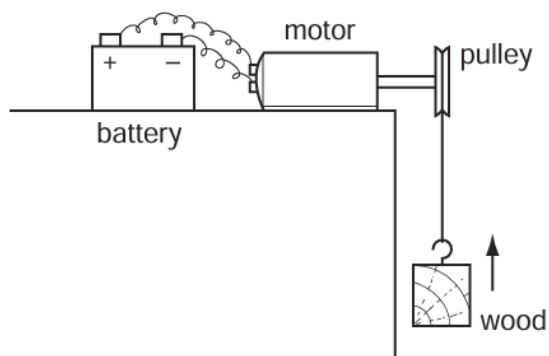
The efficiency of an electrical generator is 65 %.

Which useful output can be expected if the energy input to the generator is 12 kJ?

- A** 4.2 kJ
- B** 7.8 kJ
- C** 19 kJ
- D** 780 kJ

130. M/J/07/Q9

The diagram shows a battery-operated motor lifting a block of wood at constant speed.



What is the overall energy change taking place?

- A** chemical $\rightarrow$ gravitational potential
- B** gravitational potential $\rightarrow$ electrical
- C** gravitational potential $\rightarrow$ kinetic
- D** kinetic $\rightarrow$ gravitational potential

131. M/J/07/Q10

Four students exercise in a gym.

Which student does the most work?

	exercise time / s	power developed / W
A	50	250
B	100	150
C	200	200
D	250	30

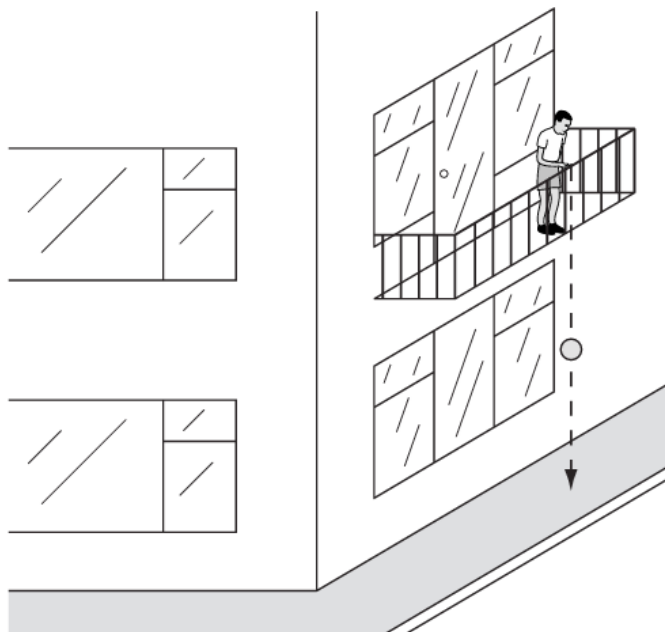
132. M/J/07/Q11

Which formula gives the efficiency of an energy conversion?

- A** efficiency = total energy input - useful energy output
- B** efficiency = useful energy output $\times$ total energy input
- C** efficiency = $\frac{\text{useful energy output}}{\text{total energy input}}$
- D** efficiency = $\frac{\text{total energy input}}{\text{useful energy output}}$

133. O/N/06/Q8

A young child holds a ball over the edge of a balcony. The ball has gravitational potential energy. The ball is then released. It falls onto a concrete path below, and bounces back up.



Which sequence represents, in the correct order, the transformations of the gravitational potential energy after the ball is released?

- A → elastic potential energy → kinetic energy → chemical potential energy
- B → elastic potential energy → kinetic energy → gravitational potential energy
- C → kinetic energy → elastic potential energy → kinetic energy
- D → kinetic energy → gravitational potential energy → kinetic energy

134. O/N/06/Q9

How much energy would be released if 1×10^{-20} kg of matter was entirely converted to energy? (The speed of light is 3×10^8 m/s.)

- A 3×10^{-12} J B 9×10^{-7} J C 4.5×10^{-4} J D 9×10^{-4} J

135. M/J/06/Q11

In a hydroelectric power station, water from a reservoir falls down a long pipe before entering the turbines. The turbines then turn the generator.

What is the **overall** energy conversion?

- A electrical energy into kinetic energy
- B electrical energy into potential energy
- C kinetic energy into chemical energy
- D potential energy into electrical energy

136. M/J/06/Q12

Which statement about fission or fusion is correct?

- A** During fission, hydrogen converts into helium and releases energy.
- B** During fission, uranium converts into daughter products and releases energy.
- C** During fusion, helium converts into hydrogen and releases energy.
- D** During fusion, uranium converts into daughter products and releases energy.

137. M/J/06/Q13

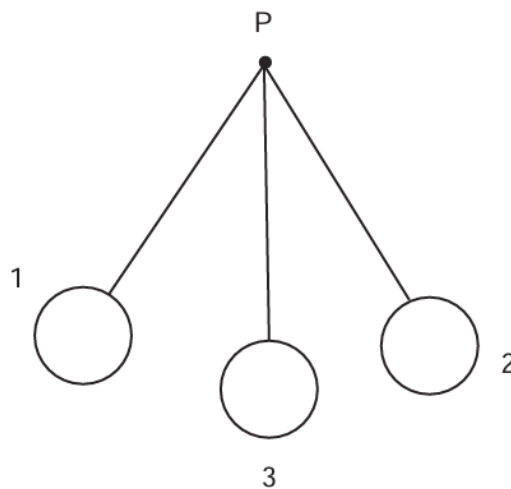
The input power to a motor is 300 W. In 20 s it lifts a load of 400 N through a height of 6.0 m.

What is the efficiency of the motor?

- A** 12 % **B** 25 % **C** 40 % **D** 75 %

138. O/N/05/Q11

A mass hangs on a string fixed at point P. It starts from position 1 and swings to the furthest position on the opposite side, position 2. It then oscillates several times with decreasing amplitude before ending at position 3.

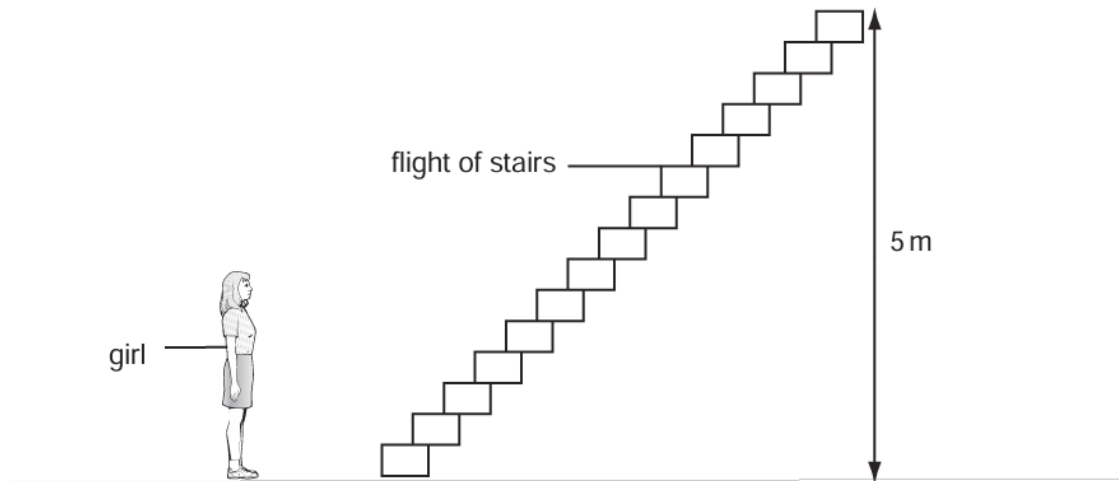


Where does the ball have the most kinetic energy?

- A** at position 1
- B** at position 2
- C** the first time at position 3
- D** the last time at position 3

139. O/N/05/Q12

A girl of weight 500 N runs up a flight of stairs in 10 seconds. The vertical height of the stairs is 5 m.



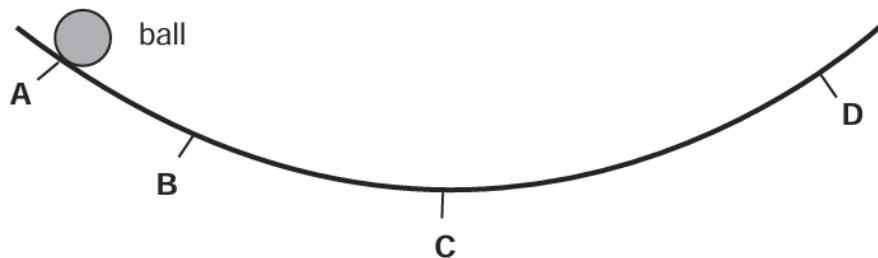
What is the average power developed by the girl?

- A** 50 W **B** 100 W **C** 250 W **D** 1000 W

140. M/J/05/Q10

The diagram shows a curved curtain rail that has a steel ball rolling on it. The ball is released at point A.

At which point does the ball have maximum kinetic energy?



141. M/J/05/Q11

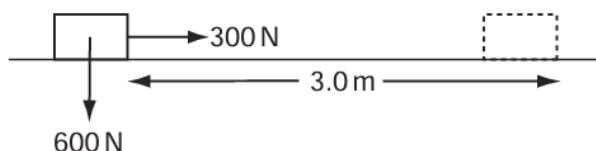
The speed of light is 3×10^8 m/s.

What is the energy equivalent of 2 kg of matter?

- A** $2 \times 3 \times 10^8$ J
B $2000 \times 3 \times 10^8$ J
C $\frac{1}{2} \times 2 \times (3 \times 10^8)^2$ J
D $2 \times (3 \times 10^8)^2$ J

142. O/N/04/Q13

When a 300 N force is applied to a box weighing 600 N, the box moves 3.0 m horizontally in 20 s.



What is the average power?

- A** 45 W **B** 90 W **C** 900 W **D** 1800 W

143. M/J/04/Q11

When a nucleus of Uranium-235 absorbs a neutron, nuclear fission occurs. In a typical reaction the total mass decreases by 3×10^{-28} kg.

Given that the speed of light c is 3×10^8 m/s, approximately how much energy is released?

- A** 9×10^{-20} J
B 2×10^{-13} J
C 3×10^{-11} J
D 3×10^{-5} J

144. M/J/04/Q12

A car is driven along a level road. The total energy input from the petrol is 60 kJ, and the car wastes 45 kJ of energy.



What is the efficiency of the car?

- A** 15 % **B** 25 % **C** 45 % **D** 75 %

145. M/J/04/Q13

Which represents the main energy changes that take place in a coal-fired power station?

- A** chemical → heat → kinetic → electrical
B chemical → heat → light → electrical
C chemical → kinetic → electrical → potential
D kinetic → heat → light → electrical

ANSWERS

1. A	48. A	97. A
2. D	49. D	98. D
3. A	50. D	99. B
4. D	51. C	100. B
5. D	52. B	101. D
6. A	53. D	102. C
7. A	54. B	103. C
8. D	55. B	104. A
9. B	56. C	105. A
10. B	57. B	106. D
11. A	58. D	107. A
12. B	59. D	108. D
13. A	60. C	109. B
14. B	61. B	110. D
15. D	62. A	111. C
16. A	63. A	112. C
17. D	64. A	113. B
18. B	65. D	114. B
19. D	66. D	115. A
20. A	67. A	116. C
21. A	68. D	117. C
22. A	69. B	118. B
23. D	70. C	119. D
24. A	71. A	120. A
25. B	72. B	121. B
26. A	73. D	122. B
27. C	74. B	123. C
28. A	75. C	124. D
29. A	76. B	125. B
30. A	77. C	126. B
31. D	78. C	127. A
32. B	79. A	128. B
33. D	80. A	129. B
34. B	81. B	130. A
35. A	82. C	131. C
36. B	83. C	132. C
37. C	84. D	133. C
38. A	85. B	134. D
39. D	86. C	135. D
40. C	87. D	136. B
41. C	88. C	137. C
42. D	89. A	138. C
43. D	90. A	139. C
44. A	91. D	140. C
45. A	92. B	141. D
46. D	93. C	142. A
47. B	94. C	143. C
	95. C	144. B
	96. A	145. A

1. O/N/23/21/Q2

A lorry (truck) is at rest on a straight, horizontal road.

A constant driving force of $5.4 \times 10^4 \text{ N}$ acts on the lorry and it accelerates forwards.

(a) Define what is meant by 'acceleration'.

.....
.....
..... [2]

(b) As the lorry moves forwards, work is done on it by the driving force.

(i) State the equation that defines work done.

Refer to the direction of the force exerted.

.....
..... [1]

(ii) Calculate the work done on the lorry by the driving force when the lorry travels a distance of 280 m.

work done = J [2]

(iii) At one point on the road, the kinetic energy of the lorry is $3.2 \times 10^6 \text{ J}$ and its speed is 12 m/s.

Calculate the mass of the lorry.

mass = kg [3]

[Total: 8]

2. M/J/23/22/Q2

Fig. 2.2 shows a rider on an electric scooter.



Fig. 2.2

The scooter contains a battery and a motor to drive the back wheel.

- (a) (i)** State the name of the energy store in the battery.

..... [1]

- (ii)** Describe, in terms of work done, the stages of energy transfer from the energy store in the battery to the kinetic energy of the scooter.

.....
.....
.....
..... [2]

- (b)** The total mass of the scooter and the rider is 70 kg.

Calculate the total kinetic energy of the rider and scooter when the scooter has a speed of 4.0 m/s.

kinetic energy = J [2]

(c) The battery is marked 'energy capacity 0.35 kilowatt-hour (kW h)'.

(i) Define what is meant by a kilowatt-hour.

.....
..... [1]

(ii) The scooter stops working because the battery is totally discharged (flat). This means that there is no more energy stored in the battery.

The battery is then recharged using a 70W power supply.

Calculate the time taken to fully recharge the battery.

time = hours [2]

[Total: 8]

3. O/N/22/21/Q2

A child's toy consists of a flexible track and a model car.

Fig. 2.1 shows a diagram of the toy.

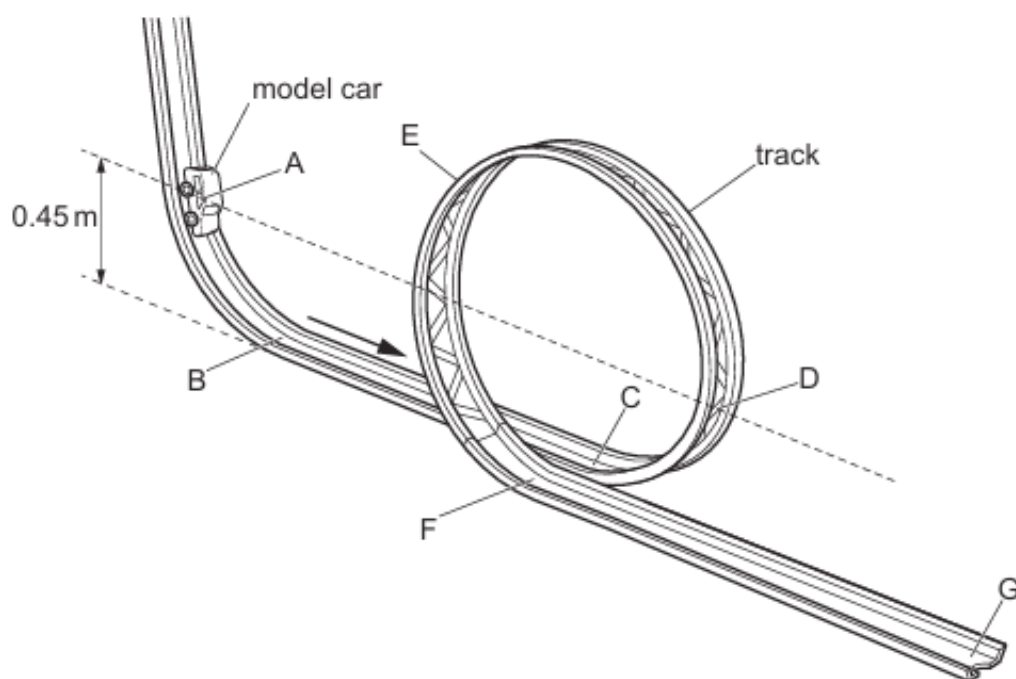


Fig. 2.1

The child first holds the car stationary at point A which is 0.45 m above the horizontal sections of track BC and FG. The mass of the car is 0.12 kg.

The child then releases the car which travels towards point B. Both air resistance and friction between the car and the track are negligible.

The gravitational field strength g is 10 N/kg.

- (a) Calculate the change in gravitational potential energy (g.p.e.) of the car as it travels from A to B.

change in g.p.e. = [2]

- (b) Calculate the speed of the car when it reaches B.

speed = [3]

- (c) After releasing it, the child expects the car to follow the track along the route ABCDEFG. In fact, the model car does not reach F.

- (i) Explain, in terms of energy, why the car does not go past D, which is also 0.45 m above the horizontal track.

.....
..... [1]

- (ii) Immediately after being released at A, the car travels to B, to C and then to D.

Describe the motion of the car after it reaches D.

.....
..... [1]

[Total: 7]

4. O/N/22/22/Q2

The foundations that support a building are long concrete cylinders that are pointed at one end. A pile-driver is a machine that forces the pointed concrete cylinders into the ground.

Fig. 2.1 shows a pile-driver.

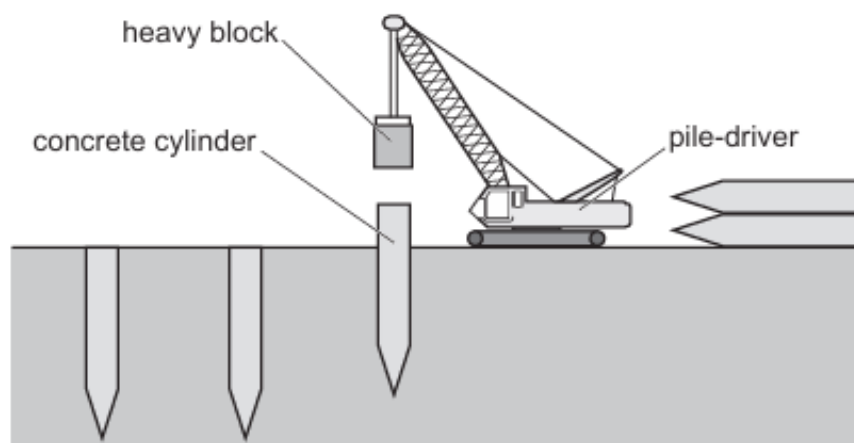


Fig. 2.1

A heavy block of mass $2.9 \times 10^4 \text{ kg}$ is lifted into the air then dropped onto the top of a concrete cylinder. This forces the cylinder into the ground.

(a) Fig. 2.2 shows the heavy block.

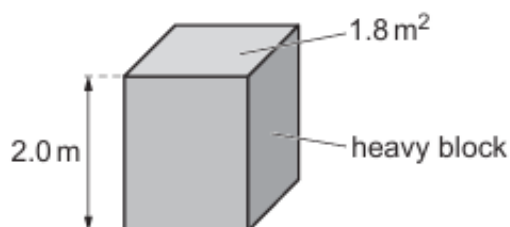


Fig. 2.2

The block is 2.0 m tall and has a cross-sectional area of 1.8 m².

Calculate the density of the material used to make the block.

density = [2]

- (b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.

The gravitational field strength g is 10 N/kg.

- (i) Calculate the gravitational potential energy gained by the block.

gravitational potential energy = [2]

- (ii) The block is then dropped from rest onto the top of the concrete cylinder.

Calculate the speed of the block just before it hits the concrete cylinder.

speed = [3]

[Total: 7]

5. M/J/22/21/Q2

Fig. 2.1 shows a wind turbine.

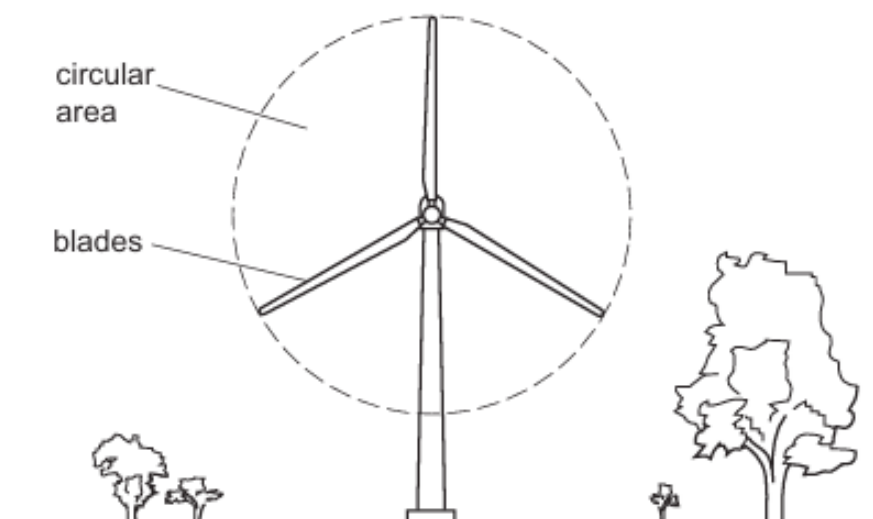


Fig. 2.1

- (a) The wind blows directly towards the turbine with a speed of 12 m/s .

In one second, $60\,000 \text{ kg}$ of air passes through the circular area swept out by the blades.

Calculate the kinetic energy of this mass of air.

kinetic energy = [3]

- (b) A wind turbine releases no carbon dioxide into the atmosphere while generating electricity. Compared with a coal-fired power station, this is an advantage of using a wind turbine.

- (i) State **one** reason why it is important to reduce the amount of carbon dioxide produced.

.....
..... [1]

- (ii) Suggest **one** advantage of using a coal-fired power station compared with a wind turbine.

.....
..... [1]

- (c) A coal-fired power station releases 0.96 kg of carbon dioxide when it generates 1.0 kWh of electrical energy.

(i) Define the kilowatt-hour (kWh).

.....

..... [1]

(ii) Calculate the mass of carbon dioxide saved when the wind turbine has a power output of 2000 kW and operates for 12 hours.

mass = [1]

(d) Wind energy is a form of renewable energy.

State the name of **one** other form of renewable energy.

..... [1]

[Total: 8]

6. O/N/21/21/Q4

In a coal-fired power station, coal is burnt and electrical energy is produced.

- (a) State the form of energy in the coal that is transferred to electrical energy.

..... [1]

- (b) (i) State the principle of the conservation of energy.

.....
.....
.....
..... [2]

- (ii) The quantity of electrical energy produced by the power station is much less than the quantity of energy in the coal that is burnt to generate the electricity.

Explain why this is so.

.....
.....
.....
..... [2]

- (c) State **one** environmental consequence of generating electricity in coal-fired power stations.

.....
..... [1]

[Total: 6]

7. M/J/21/22/Q9

(a) Fig. 9.1 represents the conversion of energy in a mobile phone.



Fig. 9.1

(i) State the form of energy stored in the battery.

..... [1]

(ii) State the useful form of output energy from the battery.

..... [1]

(iii) State **two** useful forms of output energy from the mobile phone.

..... [2]

(b) When the battery is in use, the average current is 1.3 mA.

Determine the charge that passes through the battery in a time of 2.0 minutes.

charge = [2]

- (c) A student estimates the thermal energy produced in the battery when the mobile phone is used and uses this value to estimate the efficiency of the battery.

He measures a temperature rise of 5.0°C within the battery in a period of time when the useful energy output from the battery is 5200 J .

The mass of the battery is 110 g and its specific heat capacity is $830\text{ J/(kg }^{\circ}\text{C)}$.

- (i) Calculate the thermal energy (heat) needed to raise the temperature of the battery by 5.0°C .

thermal energy = [3]

- (ii) State what is meant by *efficiency*.

.....
..... [2]

- (iii) Calculate the efficiency of the battery.

efficiency = [2]

- (iv) Suggest and explain **one** reason why the value for the efficiency calculated in (iii) is larger than the actual efficiency of the battery.

.....
.....
.....
..... [2]

[Total: 15]

8. M/J/21/21/Q8(c)

Fig. 8.1 shows a stationary horse and its rider, about to jump over two fences.

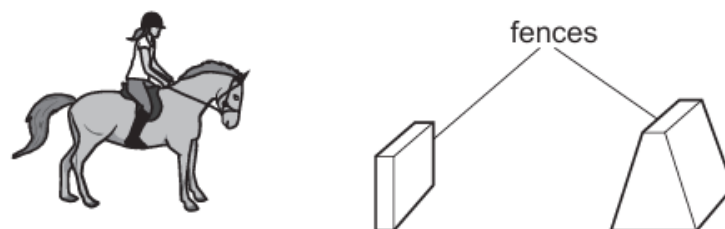


Fig. 8.1

(a) Fig. 8.2 shows a side view of the horse.

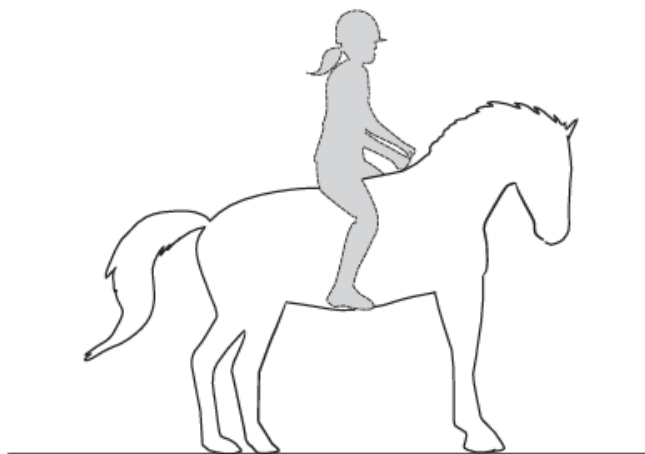


Fig. 8.2

(c) The total mass of the horse and rider is 520 kg.

(i) As they approach a fence, the horse and rider have a total kinetic energy of 4000 J.

Calculate their speed.

speed = [3]

- (ii) The centre of mass of the horse and rider is 1.4 m above the ground.

The maximum potential energy gained by the horse and rider as they jump over the fence is 3000 J.

Calculate the maximum height above the ground of the centre of mass during the jump.

The gravitational field strength $g = 10 \text{ N/kg}$.

height = [3]

9. O/N/20/22/Q2

The total mass of a wheelbarrow and its load is 90 kg. A worker pushes the wheelbarrow and load up a plank of length 2.0 m on to a platform, as shown in Fig. 2.1. The platform is at a height of 0.60 m above the ground.

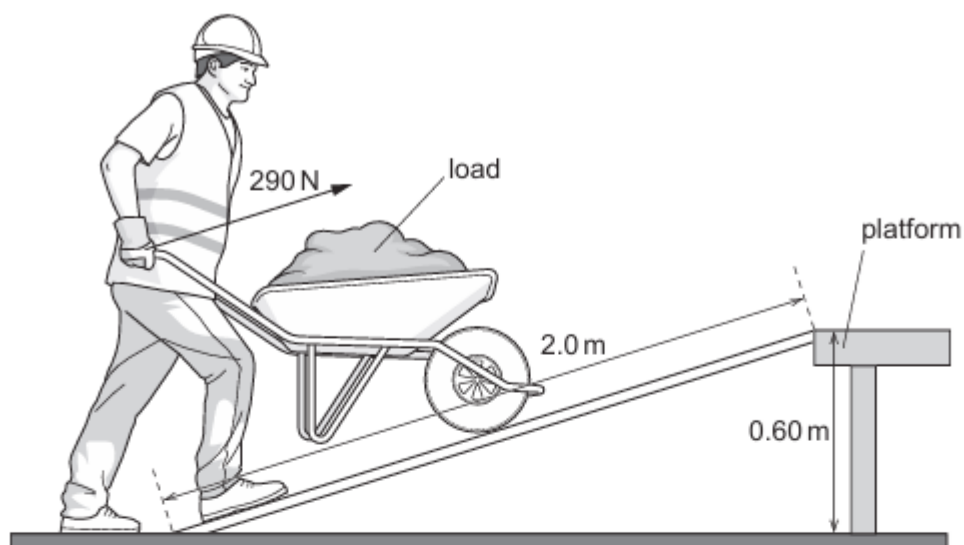


Fig. 2.1

The worker exerts a force of 290 N on the wheelbarrow in the direction in which the wheelbarrow moves.

(a) The gravitational field strength g is equal to 10 N/kg.

(i) Calculate the gravitational potential energy gained by the wheelbarrow and its load.

energy = [2]

(ii) The worker pushes the wheelbarrow 2.0 m along the plank.

Calculate the work done on the wheelbarrow by the worker.

work done = [2]

(iii) Suggest one reason why the answer to **(a)(ii)** is not equal to the answer to **(a)(i)**.

.....
..... [1]

(b) The worker finds that walking up the plank pushing the wheelbarrow is even more inefficient than the answers in (a) suggest.

(i) State what is meant by *efficiency*.

.....

..... [1]

(ii) Suggest **one** reason why this method of lifting the load onto the platform is so inefficient.

.....

..... [1]

[Total: 7]

10. M/J/20/22/Q6

The clothes iron shown in Fig. 6.1 is connected to the electrical mains.

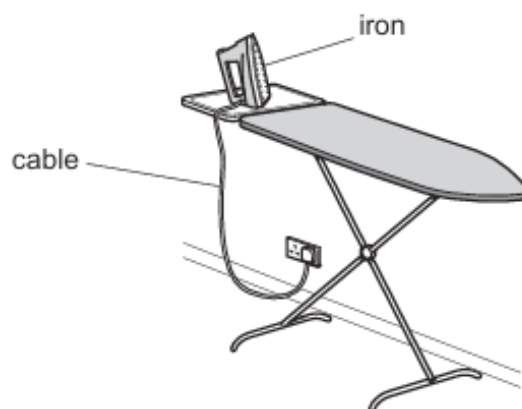


Fig. 6.1

- (a) The boxes in the left column below contain some electrical hazards. The boxes in the right column contain methods of protection from these hazards.

For each hazard, draw **one** line to the appropriate method of protection.

electrical hazard	method of protection
worn insulation on the cable to the iron	earth wire and fuse in plug correctly connected to iron
loose live wire in the iron touches its metal case	circuit breaker correctly connected in circuit
cable becomes too hot because current is too high	visual check of cable before connecting to mains

[2]

- (b) The power of the iron is 1200 W. The cost of 1 kWh of electrical energy is 20 cents (20 c).

- (i) Define the *kilowatt-hour* (kWh).

.....
 [1]

- (ii) The iron is on at full power for 20 minutes.

Calculate the cost of running the iron for this time.

cost = [2]

11. M/J/20/21/Q10

Fig. 10.1 shows a motor lifting a mass. Fig. 10.2 shows part of the circuit diagram of the connections to the motor.

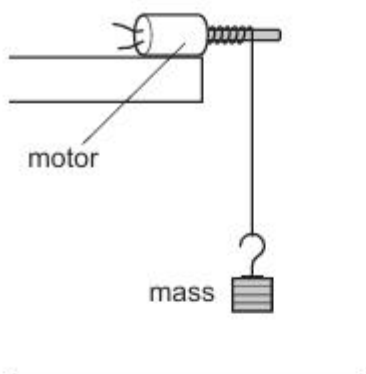


Fig. 10.1

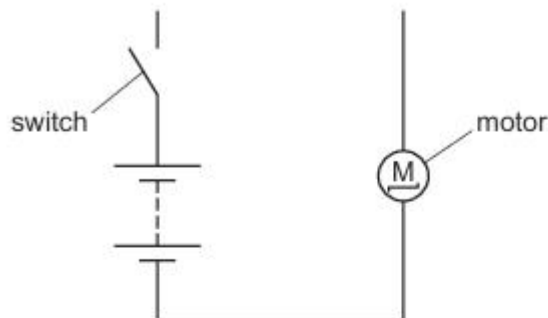


Fig. 10.2

- (a) The current in the motor is 1.5A and the voltage supplied by the battery is 8.0V.
- (i) Complete the circuit diagram in Fig. 10.2 to show an ammeter and a voltmeter in the correct positions to take these measurements while the motor is working. [2]
- (ii) The motor takes 4.0 s to lift the mass.

Calculate the electrical energy transferred to the motor in this time.

energy = [2]

- (iii) The motor lifts the 150 g mass through a height of 80 cm in the 4.0 s.

Calculate the gravitational potential energy gained by the mass.

The gravitational field strength $g = 10 \text{ N/kg}$.

gravitational potential energy = [3]

- (iv) State **two** reasons why the gravitational potential energy gained by the mass is less than the electrical energy supplied to the motor.

1.

2.

[2]

12. O/N/19/21/Q2

A small coin of mass m is initially at rest. It is dropped from a height h above ground level. The coin has a speed v as it hits the ground.

The gravitational field strength g is equal to 10 N/kg .

(a) State an expression for:

- the gravitational potential energy of the coin at a height h above ground level

gravitational potential energy =

- the kinetic energy of the coin when travelling at a speed v .

kinetic energy = [1]

(b) The coin is dropped from the top of a building of height 380 m .

Determine the speed of the coin as it hits the ground. You may ignore air resistance.

speed = [3]

(c) When air resistance is negligible, a heavier coin hits the ground at the same speed as a lighter coin when they are both dropped from the same height.

(i) Explain why.

.....
.....
..... [1]

(ii) When air resistance acts, coins of different masses do **not** hit the ground at the same speed when they are dropped from the same height.

Explain why.

.....
.....
..... [2]

13. M/J/19/21/Q3

A large spring is stretched by an athlete to increase the strength of his arms.

The athlete extends the spring, as shown in Fig. 3.1, and then releases it gently. He extends and releases the spring several times during a period of 60 s.



Fig. 3.1

- (a) State the form of energy stored in the stretched spring and the form of energy stored in the muscles of the athlete.

form of energy in spring

form of energy in muscles

[2]

- (b) Suggest what happens if he extends the spring beyond the limit of proportionality.

.....

..... [1]

- (c) During one extension of the spring, its length increases from 70 cm to 93 cm. The average force exerted by the athlete is 400 N.

- (i) Calculate the work done by the athlete in one extension.

work done = [2]

- (ii) Describe how the athlete can use the value obtained in (i) to determine his average power during the 60 s.

.....

.....

.....

..... [2]

14. M/J/18/22/Q9

Wind energy is a renewable energy source. A wind turbine and generator convert energy in the wind to electrical energy in a generator. Fig. 9.1 shows some wind turbines.

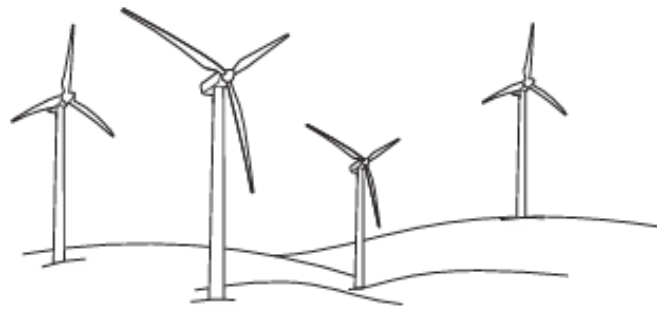


Fig. 9.1

- (a) (i) State what is meant by a *renewable energy source*.

.....
.....
.....[1]

- (ii) State the name of **one other** renewable energy source.

.....
.....[1]

- (iii) Some methods of generating electricity cause much more global warming than wind energy.

1. State **one** method of generating electricity that causes a large amount of global warming.

.....[1]

2. Describe, in outline, how the method you have chosen causes a significant amount of global warming.

.....
.....
.....[1]

- (b)** During a 30 minute period, a mass of 4.2×10^7 kg of air enters the turbine with a speed of 15 m/s.

- (i)** Calculate the kinetic energy of the air that enters the turbine in 30 minutes.

energy =[2]

- (ii)** The electrical power output of the generator connected to the turbine is 8.4×10^5 W.

Calculate the electrical energy output from the turbine in 30 minutes.

energy output =[2]

- (iii)** Calculate the efficiency of the turbine and generator in converting the kinetic energy of the air to electrical energy.

efficiency =[2]

15. M/J/18/21/Q4

A woman does work as she raises a flag by pulling downwards on a rope. This is shown in Fig. 4.1.

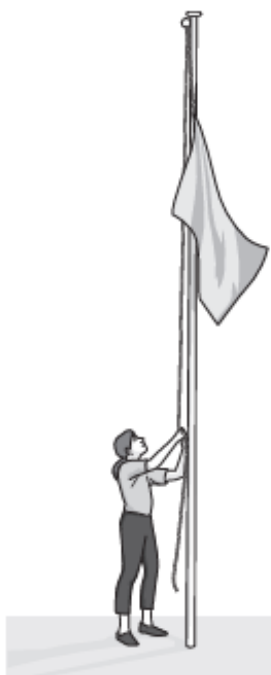


Fig. 4.1

- (a) Define what is meant by *work done*.

.....
.....[2]

- (b) The mass of the flag is 0.35 kg and the flag is raised through a height of 6.0 m.

The gravitational field strength g is 10 N/kg.

Calculate the increase in gravitational potential energy of the flag.

gravitational potential energy =[2]

- (c) Suggest and explain a reason why the work done on the rope is larger than the value calculated in (b).

.....
.....
.....
.....[2]

16. M/J/17/22/Q2

A swing is made by tying rope loosely to the branch of a tree, as shown in Fig. 2.1.

A child swings backwards and forwards several times, starting at the highest point A.

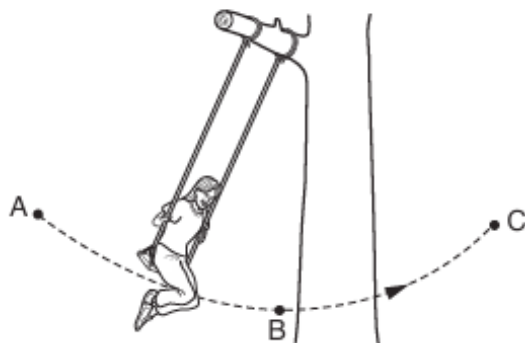


Fig. 2.1

- (a) Explain how another child can obtain an accurate measurement of the time for one complete swing.

.....
.....
.....
.....
.....[2]

- (b) When the child moves from A to B, she falls a vertical distance of 0.60 m. She loses 240 J of gravitational potential energy. The gravitational field strength g is 10 N/kg.

- (i) Calculate the mass of the child.

mass =[2]

- (ii) Suggest two reasons why her kinetic energy at B is not equal to 240 J.

1.
.....
2.
.....

[2]

17. M/J/17/21/Q9

Different energy sources are used to generate electricity.

(a) Energy sources are renewable or non-renewable.

(i) Nuclear energy is described as a non-renewable source.

Explain what is meant by a *non-renewable* energy source.

.....
[1]

(ii) Four of the energy sources used are:

hydroelectric oil geothermal wind

Write the name of these energy sources in the correct column of the table below.

non-renewable	renewable and caused by energy from the Sun	renewable and not caused by energy from the Sun

[3]

(iii) State one way in which using nuclear energy is better for the environment than using oil.

.....
[1]

(iv) State one way in which using nuclear energy is worse for the environment than using oil.

.....
[1]

(b) Fig. 9.1 is a block diagram of a power station that produces electrical energy from oil.

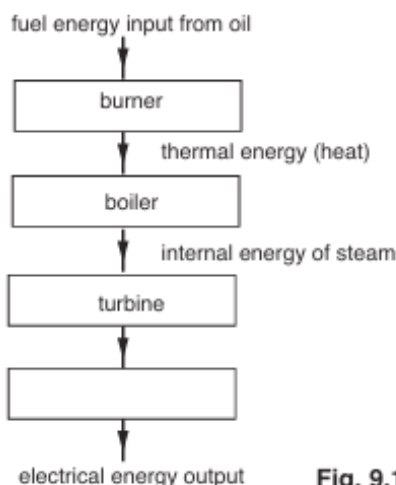


Fig. 9.1

(i) Write the name of the missing part of the power station in the empty box on Fig. 9.1. [1]

(ii) State the form of energy that the turbine possesses.

.....[1]

18. O/N/16/21/Q3

A large power station burns oil to heat water in a boiler.

- (a) State the name of the form of energy stored in the oil before it is burnt.

.....[1]

- (b) (i) State and explain whether oil is a renewable or a non-renewable energy source.

.....
.....
.....[1]

- (ii) Describe one environmental issue caused by burning oil to generate electricity in a power station.

.....
.....[1]

- (c) (i) State what is meant by *efficiency*.

.....
.....[1]

- (ii) The efficiency of the power station is 0.38 (38%). Its electrical power output is 1.9×10^9 W.

Calculate

1. the input power to the power station,

input power =[2]

2. the energy wasted in 2.0 hours.

energy wasted =[2]

19. M/J/16/22/Q2

Fig. 2.1 shows an electric motor lifting a load.

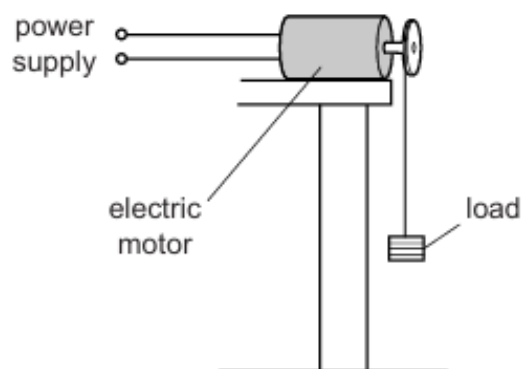


Fig. 2.1

The load, of weight 5.0 N , is raised through a vertical height of 3.5 m at a constant speed. The efficiency of the electric motor is 0.65 (65%).

- (a)** Calculate the increase in gravitational potential energy of the load.

gravitational potential energy =[2]

- (b) (i)** State the formula that relates efficiency, energy input and useful energy output.

.....
[1]

- (ii)** Calculate the energy input to the motor.

energy input =[2]

- (c)** Suggest one reason why the efficiency of the motor is less than 1.0 (100%).

.....
[1]

20. O/N/15/22/Q8

A spotlight in a television studio is at full brightness. The lamp inside the spotlight is powered by a 230 V supply and the current in the filament of the lamp is 27 A.

- (a) Calculate the power of the lamp.

power = [2]

- (b) The spotlight is kept switched on at full brightness.

- (i) Calculate the energy transformed by the lamp in 30 minutes.

energy = [1]

- (ii) The cost of using one kilowatt-hour (kWh) of electricity is 23 cents.

Calculate the cost of using the spotlight for 30 minutes.

cost = [2]

21. O/N/15/22/Q1

Water is transported to a village in a tank pulled by a tractor.

Fig. 1.1 shows the tank being pulled by a tractor.

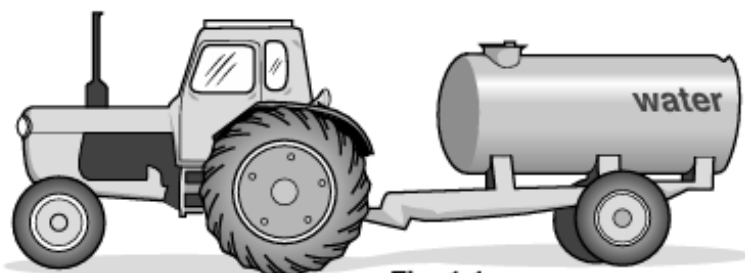


Fig. 1.1

The combined mass of the tractor and the tank is 4100 kg when the tank is empty and 6500 kg when the tank is full of water.

- (a)** The density of water is 1000 kg/m^3 .

Calculate the volume of water in the tank when it is full.

volume = [2]

- (b)** At the start of the journey, the tractor and tank accelerate from rest along a straight, horizontal road. As their speed increases, one form of energy is decreasing.

- (i)** State the name of the form of energy that is decreasing.

..... [1]

- (ii)** Explain what happens to this energy.

.....
.....
.....
..... [2]

- (c)** The village is located on a mountain at a vertical height of 850 m above the water supply. The gravitational field strength g is 10 N/kg .

Calculate the gravitational potential energy gained by the water as it is transported from the supply to the village.

energy gained = [2]

22. O/N/15/21/Q3

A lorry of mass 4.4×10^4 kg travels along a straight, horizontal road at 20 m/s.

- (a) Calculate the kinetic energy of the lorry.

kinetic energy = [3]

- (b) The lorry driver sees an obstruction ahead and applies the brakes. The lorry slows down and stops. The lorry stops a distance of 40 m from where the driver first applies the brakes.

As it slows down, the kinetic energy of the lorry decreases to zero. The kinetic energy decreases as work is done by the braking force exerted on the lorry.

- (i) State the formula that relates the work done to the braking force.

.....
.....
..... [1]

- (ii) Calculate the braking force exerted on the lorry.

force = [2]

23. M/J/13/21/Q9

Fig. 9.1 shows a conveyor belt carrying suitcases into an aeroplane.

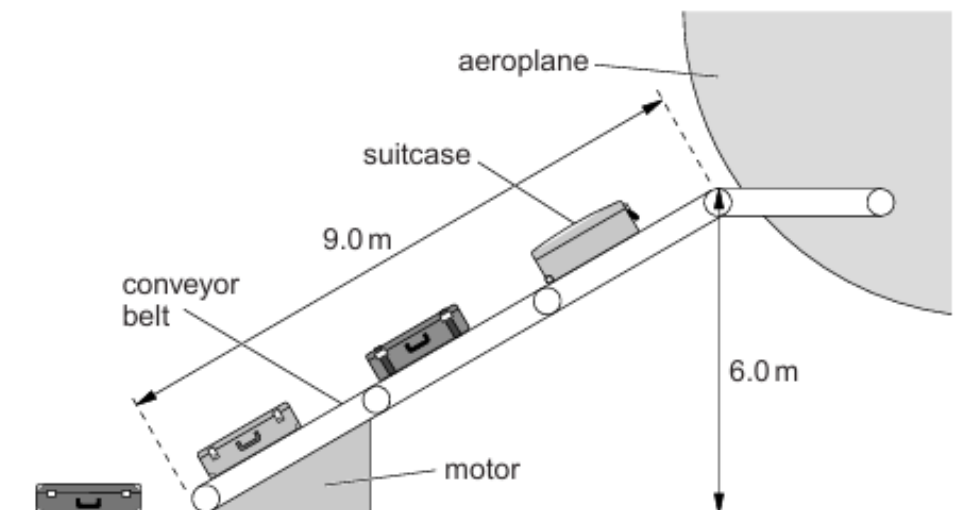


Fig. 9.1 (not to scale)

An electric motor drives the conveyor belt.

(a) A suitcase of mass 20 kg is lifted from the ground into the aeroplane.

(i) Explain what is meant by *mass* and by *weight*.

.....

 [2]

(ii) The gravitational field strength g is 10 N/kg.
 Calculate the increase in the gravitational potential energy of the suitcase.

increase in potential energy = [2]

(iii) The suitcase takes 12 s to travel 9.0 m along the conveyor belt.
 Calculate the kinetic energy of the suitcase.

kinetic energy = [3]

- (iv) The electric motor is powered by an input voltage of 230V and a current of 1.5A. Calculate the electrical energy input to the motor in 12 s.

energy =[2]

- (v) 1. State the principle of the conservation of energy.

.....
.....
.....[2]

2. Explain how this principle applies to the lifting of the suitcases into the aeroplane.

.....
.....
.....[1]

- (b) Some of the electrical energy for the motor comes from renewable energy sources.

- (i) Explain what is meant by a *renewable energy source*.

.....
.....
.....[1]

- (ii) State two renewable energy sources.

1.
2.
[2]

24. O/N/12/21/Q3

Fig. 3.1 shows an electric motor being used on a building site to lift bricks.

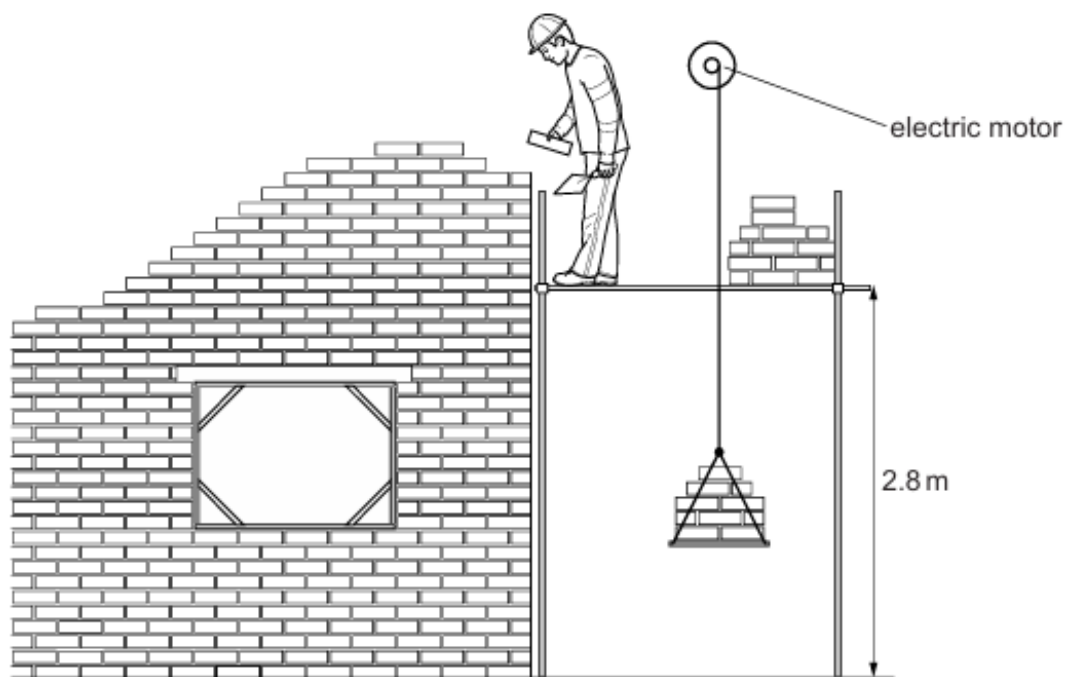


Fig. 3.1

The gravitational field strength is 10 N/kg . The motor lifts bricks of total mass 54 kg through a height of 2.8 m in 3.0 s .

(a) Calculate

(i) the work done in lifting the bricks,

work done = [2]

(ii) the useful output power of the motor.

power = [2]

(b) Suggest two reasons why the electrical power supplied to the motor is greater than its useful output power.

1.

2.

[2]

25. M/J/15/22/Q3

Two different kettles are used to heat water, as shown in Fig. 3.1.

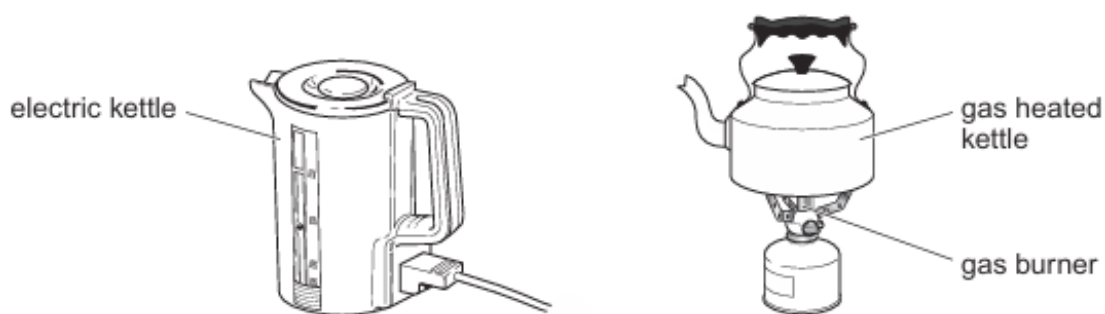


Fig. 3.1

Data for the two kettles is shown in Fig. 3.2.

	energy supplied to the kettle in one minute / J	thermal energy (heat) supplied by the kettle to the water in one minute / J
electric kettle	120 000	95 000
gas heated kettle	130 000	90 000

Fig. 3.2

(a) (i) Calculate the efficiency of the electric kettle.

efficiency = [2]

(ii) Calculate the useful power of the gas heated kettle.

power = [2]

26. O/N/14/21/Q9

Fig. 9.1 shows the dam and reservoir of a hydroelectric power station.

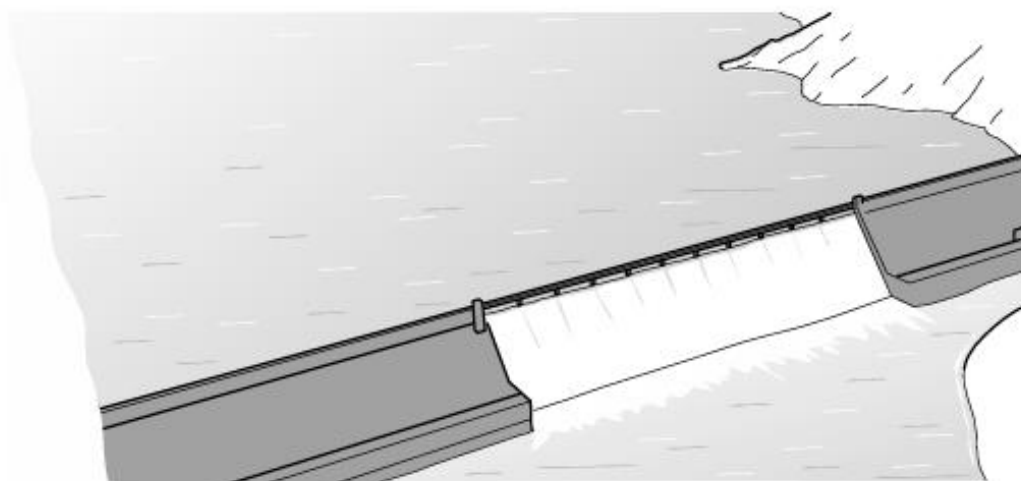


Fig. 9.1

A hydroelectric power station uses a renewable energy source.

(a) State two other renewable energy sources.

1.

2.

[2]

(b) When the power station operates at full capacity, the electrical power output is $6.8 \times 10^9 \text{ W}$.

(i) 1. Calculate the electrical energy output of the power station if it operates at full capacity for one year.

energy =[1]

2. Suggest why, in practice, the power station does not operate at full capacity throughout the year.

.....

.....[1]

- (ii) The water surface in the reservoir that supplies the hydroelectric power station is at a vertical height of 170 m above the turbines. In one hour, 1.6×10^{10} kg of water flows from the reservoir through the turbines. The gravitational field strength g is 10 N/kg.

1. Calculate the gravitational potential energy that is converted into other forms of energy in one hour.

energy =[2]

2. Calculate the efficiency of the power station when operating at full capacity.

efficiency =[2]

3. Suggest two reasons why the efficiency of the power station is less than 1.0 (100%).

1.

2.

[2]

27. O/N/12/22/Q3

A man of mass 75 kg falls from a platform high above a lake.

Fig. 3.1a shows the man tied to the platform by a long elastic rope (bungee).

Fig. 3.1b shows the man when he has fallen 20 m. After this point the rope begins to stretch.

Fig. 3.1c shows the man at 25 m below the platform where he is first stopped by the rope.

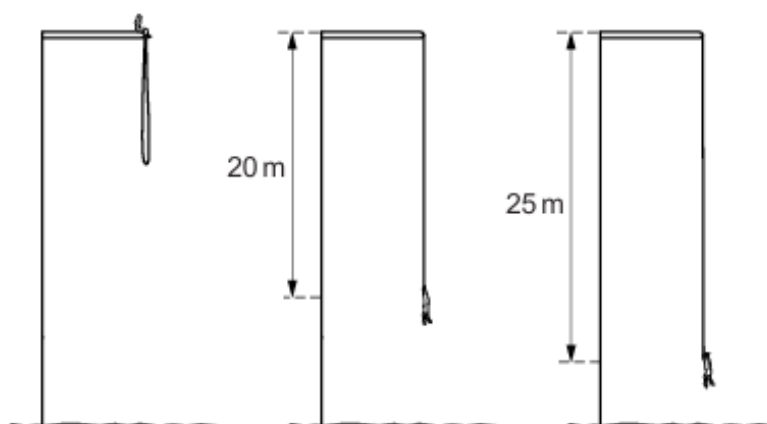


Fig. 3.1a

Fig. 3.1b

Fig. 3.1c

(a) As the man falls, his gravitational potential energy changes.

- (i)** The gravitational field strength is 10 N/kg. Calculate the change in his gravitational potential energy as he falls through 20 m.

change in potential energy =[2]

- (ii)** When he is 20 m below the platform, his kinetic energy is equal to the change in his gravitational potential energy calculated in **(i)**. Calculate his speed at this point.

speed =[3]

(b) State the energy changes that take place as he falls from 20 m to 25 m below the platform. Ignore the effect of air resistance.

.....

[3]

28. M/J/12/21/Q2

Fig. 2.1 shows a windsurfer, sail and board.

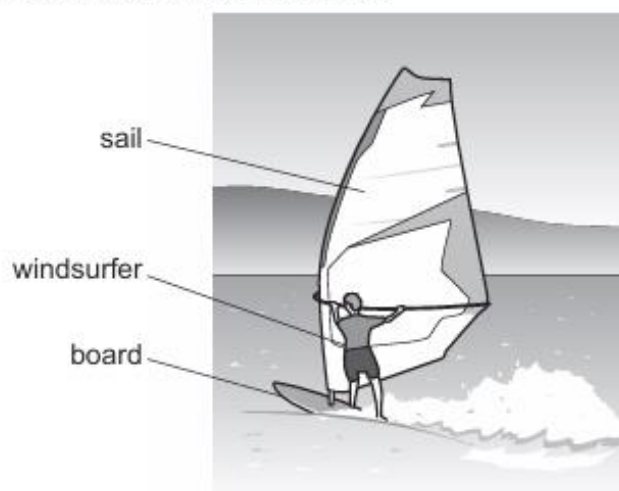


Fig. 2.1

The total mass of the windsurfer, sail and board is 90 kg. They travel at a constant speed of 5.0 m/s.

- (a) Calculate the total kinetic energy of the windsurfer, sail and board.

kinetic energy =[2]

- (b) The wind exerts a forward force on the sail.

- (i) Explain, in terms of the forces acting, why the board travels at a constant speed.

.....

[2]

- (ii) The wind does work on the sail. Explain, in terms of the conservation of energy, why the total kinetic energy does not increase.

.....
[1]

29. M/J/11/21/Q3

Fig. 3.1 shows part of a small hydroelectric system that generates electricity from moving water.

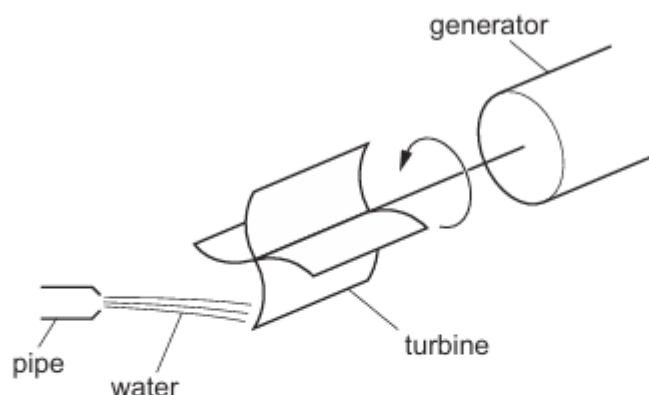


Fig. 3.1

- (a) Every minute, water with kinetic energy 14 000 J emerges from the pipe. The water turns a turbine that is connected to the generator.
Half of the kinetic energy of the water is given to the generator.

- (i) Calculate the power input to the generator.

power = [3]

- (ii) Suggest a reason why only some of the kinetic energy of the water is given to the generator.

.....
..... [1]

- (b) Hydroelectric systems use sources of renewable energy.

- (i) State what is meant by *renewable energy*.

.....
..... [1]

- (ii) State one source of **non**-renewable energy.

..... [1]

30. O/N/10/21/Q3

- (a) State the *principle of conservation of energy*.

.....
.....
..... [2]

- (b) Fig. 3.1 shows a miner using a head torch powered by a battery pack. The battery pack contains the cells that power the filament lamp in the head torch.

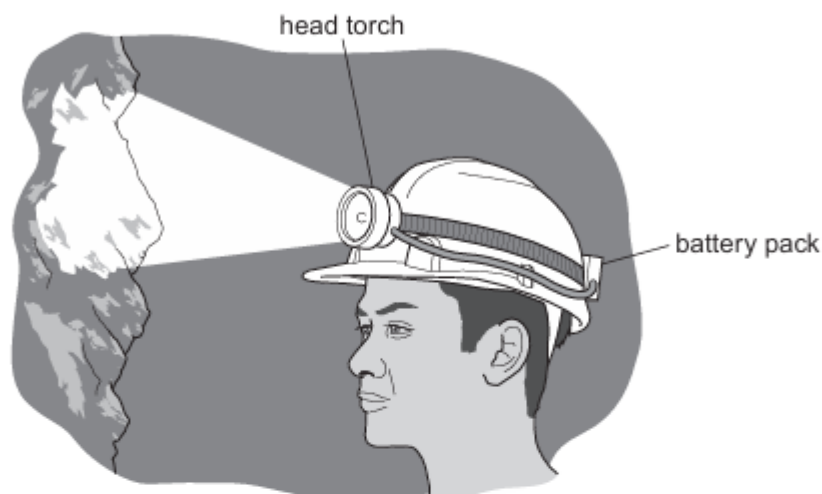


Fig. 3.1

- (i) State the main energy change that takes place in the cells.

.....
..... [2]

- (ii) State two different forms of energy that are produced in the filament lamp.

1.
2.
[2]

- (c) An identical battery pack powers a different head torch that uses light-emitting diodes (LEDs) instead of the filament lamp. The two torches are equally bright but the efficiency of the torch using the LEDs is much higher.

State how the amounts of the different forms of energy produced in the LEDs compare with the amounts produced in the filament lamp.

.....
.....
..... [2]

31. M/J/10/22/Q2

Fig. 2.1 shows a motor used to operate a lift (elevator). There is a man in the lift.

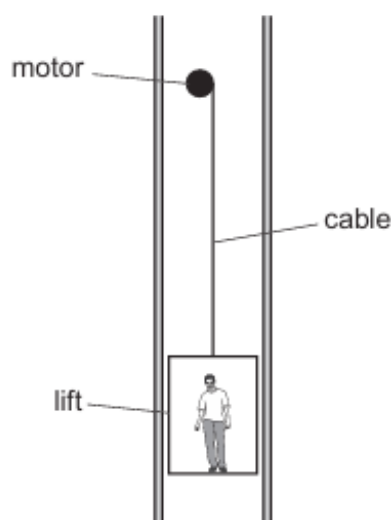


Fig. 2.1 (not to scale)

The input power to the motor is 6200 watts.

(a) Define the *watt*.

.....
 [2]

(b) The lift and the man have a total mass of 580 kg. The lift moves up a distance of 12 m in 15 s.

(i) Calculate the tension in the cable when the lift is moving at constant speed.

tension = [1]

(ii) Calculate the increase in potential energy of the lift and man.

potential energy = [2]

(iii) Calculate the efficiency of the motor.

efficiency = [2]

32. M/J/10/21/Q3

Fig. 3.1 shows a student rubbing her hands together.

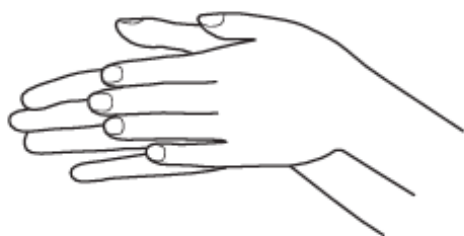


Fig. 3.1

- (a) State the main energy conversion that causes the hands to become warm.

.....
.....
.....[2]

- (b) State why the hands become even warmer if they are pressed harder together when rubbing.

.....
.....[1]

- (c) The average force used to slide one hand along the other is 1.2 N.
In each movement, one hand moves 0.080 m. The other hand remains stationary.

- (i) Calculate the number of movements needed for 2.0 J of work to be done.

number =[3]

- (ii) Each movement takes 0.20 s. Calculate the average power developed.

power =[2]

33. O/N/09/Q2

Fig. 2.1 shows a boat stationary in still water in a narrow tunnel.

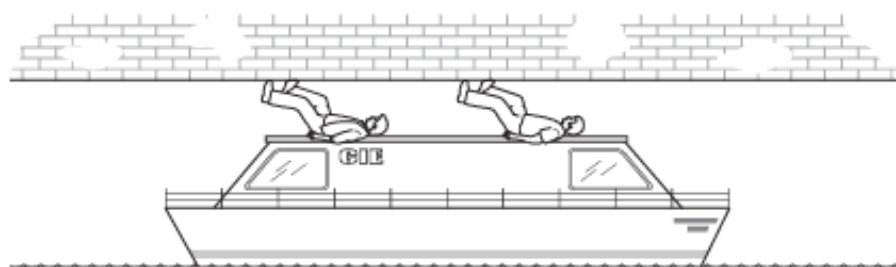


Fig. 2.1

To move the boat, two men lie on the top of it and push against the tunnel roof with their legs. They exert a total forwards force of 1680 N on the boat.

- (a) The men push the boat 50.0 m to the end of the tunnel. Calculate the work done on the boat by the men.

work done = [2]

- (b) Explain why the kinetic energy of the boat at the end of the tunnel is less than the work done on the boat by the men.

.....

 [2]

34. M/J/09/Q10

Fig. 10.1 shows the horizontal forces as a cyclist travels forwards.



Fig. 10.1

The cyclist produces the driving force that acts on the back wheel.
In this question, you may ignore any frictional force acting on the front wheel.

(a) The bicycle accelerates until a constant speed is reached.

(i) Describe how the size of the air resistance changes during this time.

.....
.....
..... [2]

(ii) Compare the sizes of the two horizontal forces when the bicycle is accelerating.

.....
..... [1]

(b) The total mass of the bicycle and the cyclist is 75 kg. At one instant, the speed of the bicycle is 4.0 m/s, the driving force is 30 N and the air resistance is 20 N.

Calculate

(i) the total kinetic energy of the bicycle and the cyclist,

.....
.....
.....
..... [3]

- (ii) the acceleration of the bicycle and the cyclist.

.....
.....
..... [2]

- (c) As the bicycle moves, energy is transmitted from the pedals to the back wheel.

Fig. 10.2 shows what happens to the energy input to the pedals.

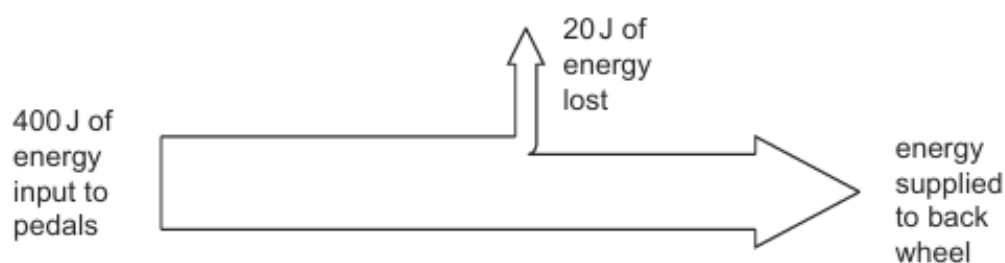


Fig. 10.2

- (i) As energy is transmitted to the back wheel, some is lost. Explain how this happens.

.....
.....
..... [2]

- (ii) Calculate the efficiency of the bicycle in transmitting energy from the pedals to the back wheel.

.....
.....
..... [2]

- (d) Some bicycles are made from low density materials. Explain why this is an advantage.

.....
.....
.....
..... [3]

ANSWERS

1.

Question	Answer	Marks
2(a)	change / increase of velocity (with time)	B1
	change / increase of velocity per unit time or rate of change / increase of velocity	B1
2(b)(i)	force $\times$ distance moved <u>in the direction of the force</u> / <u>forwards</u>	B1
2(b)(ii)	(work done $\Rightarrow$) $5.4 \times 10^4 \times 280$	C1
	1.5×10^7 (J)	A1
2(b)(iii)	$E_k = \frac{1}{2}mv^2$ or ($m \Rightarrow$) $2 \times E_k / v^2$	C1
	($m \Rightarrow$) $2 \times 3.2 \times 10^6 / 12^2$ or $2 \times 3.2 \times 10^6 / 144$	C1
	4.4×10^4 (kg)	A1

2.

Question	Answer	Marks
2(a)(i)	chemical (potential energy)	B1
2(a)(ii)	<u>electrical</u> work / (electrical) current or electrons (made to) move	B1
	mechanical) <u>work</u> is done by the <u>motor</u> / by or on the (back) <u>wheel</u>	B1
2(b)	(KE $\Rightarrow$) $\frac{1}{2}mv^2$ in any form	C1
	560 (J)	A1
2(c)(i)	(energy provided if power is) 1 kW for 1 hour / 1000 W for 1 hour	B1
2(c)(ii)	($t \Rightarrow$) E/P or $0.35 / 0.07$ or $350 / 70$	C1
	5(.0 hours)	A1

3.

Question	Answer	Marks
2(a)	(change in g.p.e.) mgh or $0.12 \times 10 \times 0.45$	C1
	0.54 J	A1
2(b)	k.e. $= \frac{1}{2}mv^2$ or $0.54 = \frac{1}{2} \times 0.12 \times v^2$	C1
	($v \Rightarrow$) $\sqrt{2gh}$ or $\sqrt{2 \times 10 \times 0.45}$ or $v^2 = 2 \times 0.54 / 0.12$ or $v^2 = 9$	C1
	3.0 m / s	A1
2(c)(i)	not enough (kinetic) energy (at C) to travel beyond D / to top of circle or more g.p.e. needed originally to travel beyond D / to top of circle or g.p.e. gained cannot be greater than original g.p.e. (transferred to k.e)	B1
2(c)(ii)	it travels back down to C and returns to A	B1

4.

Question	Answer	Marks
2(a)	$(\rho =) m / V$ or $(\rho =) m / hA$ or $2.9 \times 10^4 / 3.6$ or $2.9 \times 10^4 / (2.0 \times 1.8)$	C1
	$8.1 \times 10^3 \text{ kg / m}^3$	A1
2(b)(i)	(g.p.e. =) $mg\Delta h$ or $2.9 \times 10^4 \times 10 \times 0.80$	C1
	$2.3 \times 10^5 \text{ J}$	A1
2(b)(ii)	k.e. / $\frac{1}{2}mv^2 = \text{g.p.e.} / mg\Delta h / 2.3 \times 10^5$ or $(v =) \sqrt{2gDh}$ or $\sqrt{2 \times 10 \times 0.80}$	C1
	$(v =) \sqrt{2gDh}$ or $\sqrt{2 \times 10 \times 0.80}$	C1
	4.0 m / s	A1

5.

Question	Answer	Marks
2(a)	(K.E. =) $\frac{1}{2}mv^2$ numerical or algebraic	C1
	$0.5 \times 60\,000 \times 12^2$	C1
	4 300 000 J	A1
2(b)(i)	reduces global warming	B1
2(b)(ii)	wind not always blowing or wind not reliable / not available in some places or smaller land area (for same power) or less harm to birds or less visual effect or no interference with tv / radar signals	B1
2(c)(i)	(amount of energy of a device of power) 1 kW used for 1 hour	B1
2(c)(ii)	23 000 kg	B1
2(d)	solar, tidal, geothermal, hydro(electric), biomass, wave	B1

6.

Question	Answer	Marks
4(a)	chemical (energy)	B1
4(b)(i)	energy cannot be created and energy cannot be destroyed	B1
	energy can be transferred from one form into another	B1
4(b)(ii)	some chemical energy is transferred to thermal energy (sound and light)	B1
	thermal energy lost to air / water / buildings / equipment	B1
4(c)	any one from: carbon dioxide produced; carbon monoxide produced; global warming; greenhouse gases nitrous oxide produced; sulfur dioxide produced; acid rain; rise in sea-level	B1

7.

Question	Answer	Marks
9(a)(i)	chemical	B1
9(a)(ii)	electrical	B1
9(a)(iii)	ANY 2 from - light - sound - radio / microwave / electromagnetic (waves)	B2
9(b)	(Q =) It in any form	C1
	0.16 C	A1
9(c)(i)	($E =$) mcT numerical or algebraic in any form	C1
	$0.11 \times 830 \times 5$	C1
	460 J	A1
9(c)(ii)	energy output / energy input	C1
	useful energy output / (total) energy input	A1
9(c)(iii)	(energy input =) 460 + 5200 or 5660 (J) seen or 5200 + (energy in 9(c)(i) + 5200)	C1
	92% or 0.92	A1
9(c)(iv)	(thermal) energy <u>lost</u> / heat <u>lost</u> (from battery to environment)	B1
	energy input larger than calculated / 5660 (J) or temperature rise should be larger / is smaller than it should be or thermal energy / heat produced larger than 460 J / calculated	B1

8.

8(c)(i)	K.E. = $\frac{1}{2}mv^2$ in any form algebraic or numerical	C1
	$v^2 = 2 \times 4000 / 520$ or $v = \sqrt{\frac{2 \times 4000}{520}}$	C1
	3.9 m / s	A1
8(c)(ii)	($h =$) P.E. / mg in any form algebraic or numerical	C1
	$3000 / (10 \times 520)$ or 0.58 m	C1
	2.0 m	A1

9.

Question	Answer	Marks
2(a)(i)	(g.p.e. =) mgh or $90 \times 10 \times 0.60$	C1
	540 J	A1
2(a)(ii)	(W.D. =) Fx or 290×2.0	C1
	580 J	A1
2(a)(ii)	friction (between wheel and axle) or friction (between tyre / wheel and plank) or (work done against) friction	B1
2(b)(i)	$\frac{\text{useful energy output}}{\text{(total) energy input}} (\times 100\%)$ or $\frac{\text{useful power output}}{\text{(total) power input}} (\times 100\%)$	B1
2(b)(ii)	wheelbarrow / worker has to be lifted up ignore references to friction / air resistance	B1

10.

Question	Answer	Marks
6(a)	worn insulation → visual check loose wire in live → earth wire cable too hot → circuit breaker ALL three correct B2 one or two correct B1	B2
6(b)(i)	energy used by (a device of power) 1 kW for 1 hour	B1
6(b)(ii)	(number of kW h =) $1.2 \times 20 / 60$ or 0.4 seen	C1
	8 c	A1

11.

Question	Answer	Marks
10(a)(i)	ammeter in series connecting switch with motor	B1
	voltmeter across motor or battery	B1
10(a)(ii)	(E =) VI in any form numerical or algebraic or $P = VI$ and $P = E / t$ in any form	C1
	48 J	A1
10(a)(iii)	(E = mgh) in any form numerical or algebraic	C1
	$0.15 \times 0.8(0) \times 10$	C1
	1.2 J	A1
10(a)(iv)	ANY two of - motor / mass has kinetic energy - heat generated in motor / wires - heat generated in air due to drag / resistance <u>work done</u> to lift string	B2

12.

Question	Answer	Marks
2(a)	mgh and $\frac{1}{2}mv^2$	B1
2(b)	$\frac{1}{2}mv^2 = mgh$ or $v^2 = 2gh$ or $v^2 = 2 \times 10 \times 380$ or $v^2 = 7600$	C1
	$(v =) \sqrt{2gh}$ or $\sqrt{2 \times 10 \times 380}$	C1
	87 m / s	A1
2(c)(i)	both the g.p.e. and the k.e. are proportional to mass or mass cancels out in the equation or they accelerate at the same rate / g	B1
2(c)(ii)	air resistance (force) is a smaller proportion of weight (for heavier coin)	B1
	acceleration of heavier coin greater / less affected by air resistance	B1

13.

Question	Answer	Marks
3(a)	<i>in spring:</i> elastic or strain (potential energy)	B1
	<i>in muscles:</i> chemical (potential energy)	B1
3(b)	spring extends a different distance for the same (extra) load or load is no longer proportional to extension or becomes easier to pull	B1
3(c)(i)	(W =) Fd numerical or algebraic; 400×0.23	C1
	92 J	A1
3(c)(ii)	measure / determine the no. of times he extends the spring in 60 s	B1
	power = (total) work / time explained	B1

14.

- (a) (i) will not run out **or** infinite **or** being replaced **B1**
(ii) solar / Sun, tidal, geothermal, biomass, hydroelectric, **B1**
water waves
(iii) 1 (burning) use of a fossil fuel, e.g. coal, oil, natural gas etc. **B1**
2 produces carbon dioxide _____ **B1**
or greenhouse gases (absorb infra-red from Earth)
(b) (i) (E=) $\frac{1}{2}mv^2$ seen in any form algebraic or numerical **C1**
 $4.7 \times 10^9 \text{ J}$ **A1**
(ii) (E=)Pt seen in any form algebraic or numerical **C1**
 $1.5 \times 10^9 \text{ J}$ **A1**
(iii) (efficiency=) (energy) output / (energy) input in any form **C1**
0.32 **or** 32% **A1**

15.

- (a) force $\times$ distance (moved) **C1**
force $\times$ distance moved in direction of force **A1**
(b) mgh in any form numerical or algebraic **C1**
21 J **A1**
(c) larger force is needed (by woman / on rope) **B1**
(because of either) **B1**
 - friction / heat produced
 - weight of rope / her arms
 - flag / rope gains K.E.
 - flag accelerates / gains speed

- 16.**
- (a) use of stopwatch or electronic timer **B1**
time at least 5 swings **and** divide by number of swings **B1**
or use of fiducial mark
or definition of one swing clear e.g. A to C to A or from A and back to A
- (b) (i) $(m=)$ P.E. / gh **or** $240 / 10 \times 0.6$ **C1**
40 kg **A1**
- (ii) resistance **or** friction (with air or rope and tree) **B1**
heat produced / work done (in / against air or friction) **B1**
or effect of wind
or work done by arms / legs
- 17.**
- (a) (i) not being replaced **or** will run out **B1**
(ii) only oil in 1st column **B1**
only wind and hydroelectric in 2nd column **B1**
geothermal in 3rd column **B1**
- (iii) less greenhouse gases / global warming / acid rain / **B1**
or toxic gases / oil spills **and** how they affect a named organism / ecosystem
- (iv) nuclear waste / radioactive waste causes (storage) problems **or**
explosion / melt down / leak emits radioactivity **B1**
- (b) (i) generator (and transformer) **B1**
(ii) kinetic (energy) **B1**
- 18.**
- (a) chemical (potential energy) **B1**
- (b) (i) non-renewable **and** oil / it is not replaced / will run out **B1**
(ii) acid rain **or** produces CO_2 **or** warms lakes / rivers / sea **or** global warming **B1**
or greenhouse effect
- (c) (i) useful energy output / (total) energy input **or** power for energy twice **B1**
- (ii) **1** $1.9 \times 10^9 / 0.38$ **or** $1.9 \times 10^9 \times 100 / 38$ **C1**
 $5.0 \times 10^9 \text{ W}$ **A1**
- 2** $(E =)Pt$ **or** $0.62 \times 5.0 \times 10^9 \times 2.0 (\times 3600)$ **or** $(5.0 - 1.9) \times 10^9$ etc. **C1**
 $2.2 \times 10^{13} \text{ J}$ **A1**

19.		
(a)	(PE =) mgh or Fd or 5×3.5 17.5 J or 17 J or 18 J	C1 A1
(b) (i)	(efficiency = useful) energy output/energy input in any form but all three quantities must be mentioned if efficiency is not the subject of the equation	B1
(ii)	17.5/0.65 or 17.5/65 or $0.65/65 = (a)/\text{energy input}$ 26.9 J or 27 J	C1 A1
(c)	due to friction (in bearings of motor) or due to (electrical) resistance (in motor) or air resistance acts or thermal energy/heat produced/lost (in resistance of motor/due to friction)	B1
20.		
(a)	($P =$) VI or 230×27 6200/6210 W or 6.21/6.2 kW	C1 A1
(b) (i)	$1.1/1.12/1.1178 \times 10^7$ J	B1
(ii)	$6.21 \times 0.5 \times 23$ or $6.21 \times 30 \times 23$ or $3.1/3.105 \times 23$ 71/71.3/71.4/71.415 c or \$ 0.71/0.714/0.71415	C1 A1
21.		
(a)	($V =$) m/ρ or $10\,600/6500/4100/2400 \div 1000$ or 6.5 or 4.1 $2.4 \text{ m}^3/2.4 \times 10^6 \text{ cm}^3$	C1 A1
(b) (i)	fuel/chemical (potential energy)	B1
(ii)	some to heat/thermal (energy) some to kinetic (energy of air or tractor)	B1 B1
(c)	($GPE =$) mgh or $2400 \times 10 \times 850$ $2.0/2.04 \times 10^7$ J	C1 A1
22.		
(a)	(k.e. =) $\frac{1}{2}mv^2$ $\frac{1}{2} \times 4.4 \times 10^4 \times 20^2$ 8.8×10^6 J	C1 C1 A1
(b) (i)	$WD = F \times x$ (ii) or force $\times$ distance (parallel to/in direction of force)	B1
(ii)	$8.8 \times 10^6/40$ or $a = (-)5.0 \text{ (m/s}^2\text{)}$ or $t = 4.0 \text{ (s)}$ seen 2.2×10^5 N	C1 A1

23.

- (a) (i) mass is the amount of matter/substance **or** to resist (change in) motion **or** (measurement of) inertia B1
weight is the pull/force of gravity **or** pull of Earth B1
accept mg where g stated as 10 (N/kg) or grav. field strength or acc. due to gravity
ignore mg where g is gravity or grav. force or undefined
- (ii) mgh algebraic or numerical C1
1200 J A1
- (iii) $\frac{1}{2}mv^2$ algebraic or numerical C1
(speed =) 9/12 **or** 0.75 seen A1
5.6(25) J A1
- (iv) ($E=$) VIt algebraic or numerical C1
4100 J **or** 4140 J A1
- (v) 1. energy can neither be created or destroyed/lost B1
(but) may change form / be transferred B1
2. electrical energy changes to P.E. (and K.E. and heat/work against friction; **ign.** mechanical energy) B1
- (b) (i) will not run out **or** infinite **or** being replaced (**allow** does not finish/always available)
(**ign.** cannot be reused/recycled) B1
- (ii) wind, tidal, solar/Sun, geothermal, hydroelectric, biomass, waves, wood (**not** nuclear) B2
(**allow** biogas/biofuel e.g. cane into petrol, dung into gas etc.) (**ign.** tidal waves)

24.

- (a) (i) ($WD =$) mgh **or** $54 \times 10 \times 2.8$ C1
1500/1510/1512 J A1
- (ii) ($P =$) WD/t or E/t **or** 1500/3 **or** 1510/3 **or** 1512/3 C1
500/503/504 W A1
- (b) any **two** of:
also lifting board/rope
heat in motor/wires/cable
friction with something named e.g. axle/spindle/air B2

- 25.
- (a) (i) (efficiency =) useful energy ÷ input energy C1
or $95\,000/120\,000 (\times 100)$
0.79(17) or 79(.17)% A1
- (ii) ($P=$) energy/time or $90\,000/60$ C1
1500 W A1
- 26.
- (a) any **two** from:
biomass/wood; geothermal power; solar power; tidal power; wave power; wind power B2
- (b) (i) 1. $2.1(4) \times 10^{17}$ J (**allow** $2.1(5) \times 10^{17}$ J if candidate uses 365.24/5) B1
2. any **one** from: not enough water (to maintain maximum flow); rainfall varies (during the year); periods of low demand B1
- (ii) 1. (GPE =) mgh or $1.6 \times 10^{10} \times 10 \times 170$ C1
 $2.7(2) \times 10^{13}$ J A1
2. $2.7(2) \times 10^{13}/3600$ or $6.8 \times 10^9 \times 3600$ or $6.8 \times 10^9/7.5(55) \times 10^9$ or
 $2.4(48) \times 10^{13}/2.7(2) \times 10^{13}$ C1
0.90 or 90% A1
3. any **two** from:
friction (of water) with pipe/turbine/; viscosity of water; friction at bearings; resistance/heat in the wires; KE of water leaving turbine B2
- 27.
- (a) (i) (PE =) mgh or $75 \times 10 \times 20$ C1
 1.5×10^4 J A1
- (ii) $\frac{1}{2}mv^2$ or $\frac{1}{2}75v^2$ C1
 $v^2 = 400$ (if this is seen it scores the first 2 marks) C1
 $v = 20$ m/s A1
- (b) (G)PE at start B1
KE at start B1
to elastic/strain/clear equivalent /EPE at end
(not stretch energy; any intermediate energy –1) B1
- 28.
- (a) (K.E.) = $\frac{1}{2}mv^2$; $\frac{1}{2} \times 90 \times 5^2$ C1
1125 J or 1100 J A1
- (b) (i) no resultant force; forwards force = backwards force/drag/friction (ignore air resistance) B1
water resistance/**water** drag mentioned/**water** friction OR sail exerts force on board B1
- (ii) heat produced OR equal to work done against backwards force/drag/friction B1

29.		
(a)	(i) 7000 (J) seen or 50% used somewhere ($P = E/t$ // 14000/t // 7000/t seen // 7000 J/minute // 420 000 J/hour 120 W	C1 C1 A1
	(ii) water after hitting turbine still moves // has KE/energy/velocity // energy lost due to friction // friction and location // heat/internal energy and location // water misses turbine	B1
(b)	(i) can be replaced/made // will not run out	B1
	(ii) coal, oil, gas, peat, nuclear, uranium (not solar)	B1
30.		
(a)	energy cannot be created/destroyed (nb. only one required) energy cannot be destroyed or created (i.e. the other one as well) or (merely) transformed or total energy in an isolated system is constant	B1 B1 B1
(b)	(i) chemical (potential) at beginning to electrical (and heat) at end others present: max 1	B1 B1
	(ii) light heat/thermal/internal others present: max 1	B1 B1
(c)	less heat; same light or less chemical/electrical; less heat or less chemical/electrical; same light	B2 B2 B2
31.		
(a)	energy/time 1 joule in one second	C1 A1
(b)	(i) 5800 N or 5684 N or 5700 N	B1
	(ii) mgh algebraic, words or numerical (i.e. $580 \times 10 \times 12$) 69600 J or 70000 J or 68208 J or 68000 J	C1 A1
	(iii) (efficiency =) output power or energy/input power or energy algebraic or numerical or 93000 seen or 4640 seen 0.75 or 75% (accept 0.748) e.c.f. from (ii)	C1 A1
32.		
(a)	chemical energy to or K.E to heat/thermal energy/internal energy – at end	B1 B1
(b)	friction/resistive force increases	B1
(c)	(i) work = force $\times$ distance in words, number or symbols 1.2×0.08 or 0.096 or 20 or 0.208 seen 20.8 or 21	C1 C1 A1
	(ii) power = work/time or energy/time, numerical or algebraic – may use values in (i) accept 2/0.2 or 10 W 0.48 W (ecf (i) including power of ten error in (i))	C1 A1

- 33.
- (a) (WD =) Fx or 1680×50 C1
84 000 J A1
- (b) friction/drag/resistance of water/air B1
work done against friction/resistance/drag or energy lost due to friction/resistance/drag or energy lost as heat/internal/thermal B1
- 34.
- (a) (i) air resistance increases (as speed increases) B1
(at constant speed) becomes equal to driving force/applied force etc. B1
- (ii) driving force (forward force) larger (than air resistance/backwards force) B1
- (b) (i) ($E =$) $\frac{1}{2} mv^2$ algebraic formula C1
 $\frac{1}{2} \times 75 \times 4^2$ C1
600 J A1
- (ii) ($a =$) F/m algebraic seen or 10 (N) used as force C1
 0.13 m/s^2 A1
- (c) (i) friction (in chain/axles) or rubbing of surfaces B1
heat or thermal energy produced B1
- (ii) (efficiency = useful) energy output/energy input algebraic or numerical or 380 seen C1
0.95 or 95% A1
- (d) lower mass/weight of cycle B1
less force needed same acceleration/get up hill/to stop cycle/lift cycle
greater acceleration/easier to acc. for same force or $F = ma$ quoted
less energy/work (input) to go uphill/due to less friction/ $\frac{1}{2} mv^2$ less
less kinetic energy less stopping distance/less force to stop
more efficient/less energy wasted going uphill/less friction
less friction easier turn handlebars/higher (top) speed
less pressure (on ground) sinks less into ground M1A1